Ecuaciones Diferenciales I-2026 501-502	Experiencia de Aprendizaje: EDO de Primer Orden.
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A continuación, encontrará una serie de actividades que sirven para reforzar, comprender, aplicar y medir los aprendizajes que debe alcanzar al concluir el CADI de Ecuaciones Diferenciales. Estas actividades se deben cargar en el Campo Multidimensional de Aprendizaje CMA (o aula virtual) de manera individual, de forma escrita, en formato pdf, y utilizando un editor de textos científicos.

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Repaso:

La intención de las siguientes preguntas es repasar y recordar algunos conceptos básicos que deben tener en cuenta para iniciar este curso.

Pregunta 1. ¿Qué es y para qué sirve un factor integrante?

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Pregunta 2. ¿Qué es un modelo matemático, reconoce modelos matemáticos que se expresen como ecuaciones diferenciales ordinarias?

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Pregunta 3. ¿Recuerda el nombre y los métodos analíticos para resolver ecuaciones diferenciales ordinarias de primer orden, que se repasaron en la dimensión aula?

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Pregunta 4. ¿Qué es un campo escalar y qué es un campo vectorial?

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Pregunta 5. ¿Qué son curvas de nivel?

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Pregunta 6. ¿Qué es y qué significa el vector gradiente, cuál es su relación con las curvas de nivel?

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Pregunta 7. ¿La función $f(x, y) = x^2 + y^2$ tiene un máximo o mínimo global?

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Review

Question 8. Does the function $f(x, y) = x^2 - y^2$ have a global maximum or minimum?

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1.Draw a direction field for the given differential equation.

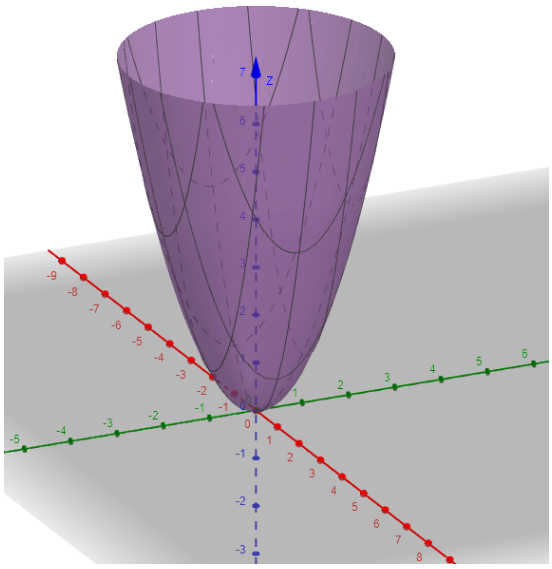

Figura 1: Graphic of the function to analyze

Checking for Global Maximum: Fix $y = 0$. Then:

$$f(x, 0) = x^2 \to +\infty \quad \text{as } x \to +\infty$$

So f is unbounded above $\Rightarrow$ **no global maximum**.

Checking for Global Minimum: Fix $x = 0$. Then:

$$f(0, y) = -y^2 \to -\infty \quad \text{as } y \to +\infty$$

So f is unbounded below $\Rightarrow$ **no global minimum**.

Critical Point Analysis: Setting the partial derivatives equal to zero:

$$\frac{\partial f}{\partial x} = 2x = 0 \implies x = 0, \qquad \frac{\partial f}{\partial y} = -2y = 0 \implies y = 0$$

The only critical point is $(0,0)$. Applying the second derivative test:

$$D = f_{xx}\, f_{yy} - (f_{xy})^2 = (2)(-2) - 0 = -4 < 0$$

Since $D < 0$, the point $(0,0)$ is a **saddle point**, not a local extremum.
Final answer:

The function $f(x,y) = x^2 - y^2$ has **no global maximum** and **no global minimum**. The unique critical point $(0,0)$ is a saddle point.

Question 9. Let $f(x,y) = xy$, $c(t) = (e^t, \cos(t))$ and $r(t) = f(c(t))$; compute and interpret dr/dt.

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First, we find $r(t)$ explicitly:

$$r(t) = f(c(t)) = f(e^t, \cos(t)) = e^t \cos(t)$$

To compute $\frac{dr}{dt}$, we apply the **Chain Rule**:

$$\frac{dr}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt}$$

We compute each part:

- $\frac{\partial f}{\partial x} = y = \cos(t)$
- $\frac{\partial f}{\partial y} = x = e^t$
- $\frac{dx}{dt} = \frac{d}{dt}(e^t) = e^t$
- $\frac{dy}{dt} = \frac{d}{dt}(\cos(t)) = -\sin(t)$

Therefore:

$$\frac{dr}{dt} = \cos(t) \cdot e^t + e^t \cdot (-\sin(t)) = e^t \cos(t) - e^t \sin(t) = e^t(\cos(t) - \sin(t))$$

Interpretation: $\frac{dr}{dt}$ represents the instantaneous rate of change of the composite function $r(t) = e^t \cos(t)$ with respect to t. It measures how fast the product $f(x,y) = xy$ changes as the point (x,y) moves along the curve $c(t) = (e^t, \cos(t))$ in the xy-plane.
Final answer:

$$\frac{dr}{dt} = e^t(\cos(t) - \sin(t))$$

Question 10. What is a direction field of a first-order ODE and what is its significance?

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Definition: Consider a first-order ODE of the form:

$$\frac{dy}{dx} = f(x, y)$$

A **direction field** (or *slope field*) is a graphical representation constructed by drawing a short line segment at each point (x, y) in the plane with slope equal to $f(x, y)$.

Construction: At each sampled point (x_0, y_0), we evaluate $f(x_0, y_0)$ and draw a small segment with that slope:

$$\text{slope at } (x_0,\, y_0) \;=\; f(x_0,\, y_0)$$

Repeating this over a grid of points produces the full direction field.

Significance:

- **Qualitative analysis:** It allows us to visualize the behavior of solutions without solving the ODE explicitly.
- **Solution curves:** Any particular solution $y = y(x)$ must be tangent to the direction field at every point it passes through.
- **Equilibria:** Horizontal segments (where $f(x, y) = 0$) reveal equilibrium solutions and their stability.
- **No analytic solution needed:** It is especially useful when the ODE has no closed-form solution.

Final answer:

A direction field of $y' = f(x, y)$ is a plot of short line segments at points (x, y) each with slope $f(x, y)$. Its significance is that it provides a geometric picture of all solutions: solution curves flow tangentially along the segments, revealing behavior, equilibria, and stability without requiring an explicit solution.

Question 11. Let $f(x, y) = (x^2 + y^2)\ln(\sqrt{x^2 + y^2})$, $c(t) = (e^t, e^{-t})$ and $r(t) = f(c(t))$; compute and interpret dr/dt.

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First, we find $r(t)$ explicitly by substituting $c(t) = (e^t, e^{-t})$ into f:

$$r(t) = f(e^t, e^{-t}) = (e^{2t} + e^{-2t})\ln\left(\sqrt{e^{2t} + e^{-2t}}\right)$$

Since $\ln(\sqrt{u}) = \frac{1}{2}\ln(u)$, we simplify:

$$r(t) = \frac{1}{2}(e^{2t} + e^{-2t})\ln(e^{2t} + e^{-2t})$$

We apply the **Chain Rule**:

$$\frac{dr}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt}$$

Computing the partial derivatives of f:
Let $u = x^2 + y^2$, so $f = u\ln(\sqrt{u}) = \frac{1}{2}u\ln(u)$.

$$\frac{\partial f}{\partial x} = \frac{1}{2}\left(\ln(u)\cdot 2x + u\cdot\frac{2x}{u}\right) = x\ln(x^2+y^2) + x = x\left(\ln(x^2+y^2)+1\right)$$

$$\frac{\partial f}{\partial y} = \frac{1}{2}\left(\ln(u)\cdot 2y + u\cdot\frac{2y}{u}\right) = y\ln(x^2+y^2) + y = y\left(\ln(x^2+y^2)+1\right)$$

Computing the derivatives of the components of $c(t)$:

$$\frac{dx}{dt} = e^t, \qquad \frac{dy}{dt} = -e^{-t}$$

Evaluating at $c(t) = (e^t, e^{-t})$:

$$\left.\frac{\partial f}{\partial x}\right|_{c(t)} = e^t\left(\ln(e^{2t}+e^{-2t})+1\right), \qquad \left.\frac{\partial f}{\partial y}\right|_{c(t)} = e^{-t}\left(\ln(e^{2t}+e^{-2t})+1\right)$$

Applying the chain rule:

$$\begin{aligned}\frac{dr}{dt} &= e^t\big(\ln(e^{2t}+e^{-2t})+1\big)\cdot e^t + e^{-t}\big(\ln(e^{2t}+e^{-2t})+1\big)\cdot(-e^{-t})\\ &= e^{2t}\big(\ln(e^{2t}+e^{-2t})+1\big) - e^{-2t}\big(\ln(e^{2t}+e^{-2t})+1\big)\\ &= \big(e^{2t}-e^{-2t}\big)\big(\ln(e^{2t}+e^{-2t})+1\big)\end{aligned}$$

Interpretation: $\frac{dr}{dt}$ represents the instantaneous rate of change of f as the point (x, y) moves along the curve $c(t) = (e^t, e^{-t})$. It combines how sensitive f is to changes in each variable (via ∇f) with how fast the curve moves in each direction (via $c'(t)$). When $t > 0$ we have $e^{2t} > e^{-2t}$, so $r(t)$ is increasing; when $t < 0$ it is decreasing; and at $t = 0$ the rate is zero.
Final answer:

$$\boxed{\frac{dr}{dt} = \big(e^{2t}-e^{-2t}\big)\big(\ln(e^{2t}+e^{-2t})+1\big)}$$

Question 12. What are the solution curves of an ordinary differential equation?

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An **ordinary differential equation (ODE)** is an equation relating a function of one variable to its derivatives. A **solution** to an ODE is a function that satisfies the equation on some interval.
Solution curves are the geometric representation of these solutions in the plane. More precisely:
Given an ODE of the form $\frac{dy}{dx} = F(x, y)$, a solution curve is the graph

$$\{(x, y(x)) : x \in I\}$$

of a function $y = y(x)$ defined on some interval $I \subseteq \mathbb{R}$ that satisfies the equation That is at every point (x_0, y_0) on the curve, the slope of the tangent line equals $F(x_0, y_0)$.
Key observations:

- The general solution of an ODE contains a family of solution curves, typically parameterized by a constant C. Each value of C gives a distinct curve in the xy-plane.
- A **particular solution** (obtained by imposing an initial condition $y(x_0) = y_0$) selects exactly oe curve from this family — the unique one passing through (x_0, y_0).
- By the existence and uniqueness theorem, solution curves **do not intersect**, since through each point in the plane there passes at most one solution curve.
- The collection of all solution curves is called the **general solution** and can be visuaized using a **direction field** (or slope field) where small arrows at each point (x, y) indicate the slope $F(x, y)$ that any solution curve must have there.

Final answer:

Solution curves of an ODE $\frac{dy}{dx} = F(x, y)$ are the graphs of all functions $y = y(x)$ satisfying the equation. They form a family of non-intersecting curves in the xy-plane, each corresponding to a particular value of the integration constant C. Imposing an initial condition $y(x_0) = y_0$ selects the unique solution curve passing through the point (x_0, y_0).

Question 13. What is the slope of the tangent line to the solution of the following differential equation $\dfrac{dy}{dt} = t^2 + y^2$, when it passes through the point a.: $(1, 2)$; b.: $(0, 0)$; c.: (π, e); are the solutions of this equation always decreasing?

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Process:
The slope of the tangent line at any point (t, y) is given directly by evaluating

the right-hand side of the ODE at that point.
a. At $(1, 2)$:

$$\left.\frac{dy}{dt}\right|_{(1,2)} = 1^2 + 2^2 = 1 + 4 = 5$$

b. At $(0, 0)$:

$$\left.\frac{dy}{dt}\right|_{(0,0)} = 0^2 + 0^2 = 0$$

c. At (π, e):

$$\left.\frac{dy}{dt}\right|_{(\pi,\, e)} = \pi^2 + e^2 \approx 9{,}87 + 7{,}39 \approx 17{,}26$$

Decreasing? Since $t^2 \geq 0$ and $y^2 \geq 0$ for all real t, y, we have $\frac{dy}{dt} = t^2 + y^2 \geq 0$ everywhere, with equality only at $(0, 0)$. Therefore, the solutions are **never decreasing**; they are always non-decreasing (strictly increasing except momentarily at the origin).
Final answer:

a. Slope $= 5$ b. Slope $= 0$ c. Slope $= \pi^2 + e^2 \approx 17{,}26$

The solutions are **never decreasing**, since $t^2 + y^2 \geq 0$ for all (t, y).

Question 14. Suppose that $y(x)$ is a function defined implicitly by $G(x, y(x)) = k$, where k is a constant and G is a function from $\mathbb{R}^2$ to $\mathbb{R}$. If $y(x)$ is differentiable, verify that: $\partial G/\partial x + (\partial G/\partial y)(dy/dx) = 0$.

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Process:
Since k is a constant, we differentiate both sides with respect to x:

$$\frac{d}{dx}\big[G(x,\, y(x))\big] = \frac{d}{dx}[k] = 0.$$

The left-hand side involves a composition: G depends on x directly and also through $y(x)$. Applying the **multivariable Chain Rule**:

$$\frac{d}{dx}\big[G(x,\, y(x))\big] = \frac{\partial G}{\partial x} \cdot \underbrace{\frac{dx}{dx}}_{=1} + \frac{\partial G}{\partial y} \cdot \frac{dy}{dx}.$$

Therefore, setting this equal to zero:

$$\frac{\partial G}{\partial x} + \frac{\partial G}{\partial y}\,\frac{dy}{dx} = 0.$$

Moreover, solving for dy/dx (provided $\partial G/\partial y \neq 0$) recovers the **Implicit Function Theorem** formula:

$$\frac{dy}{dx} = -\frac{\partial G/\partial x}{\partial G/\partial y}.$$

Final answer:

Differentiating both sides of $G(x, y(x)) = k$ with respect to x and applying the multivariable Chain Rule:

$$\frac{d}{dx}\big[G(x, y(x))\big] = \frac{\partial G}{\partial x} + \frac{\partial G}{\partial y}\frac{dy}{dx} = \frac{d}{dx}[k] = 0 \qquad \blacksquare$$

Space Travel

Question 15. Space Travel. Fundamentals for a research experience during the course of the semester t is possible that the calculation of orbital trajectories allows connecting ODEs with a complex and scientifially relevant problem. This topc should articulate gravitational laws systems of second-order differential equations and numerical and computational methods. For this reason, I recommend starting the investigation here:

1. Can ordinary differential equatins be used to model analyze and approximate the orbital trajectory of a spacecraft for example in the Artemis II mission?

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Yes. The position of a spacecraft is described by a vector $\mathbf{r}(t)$, and its motion is governed by Newton's Second Law combined with Universal Gravitation:

$$\ddot{\mathbf{r}} = -\frac{\mu}{|\mathbf{r}|^3}\,\mathbf{r}, \qquad \mu = GM$$

This is a second-order ODE that expands into three scalar equations in x, y, z. By introducing velocity components (v_x, v_y, v_z), the system reduces to **6 coupled first-order ODEs** standard initial value problem where launch position and velocity serve as initial conditions.
For a mission like Artemis, the gravitational influence of both Earth and the Moon must be included, giving the **three-body problem**:

$$\ddot{\mathbf{r}} = -\frac{GM_E}{|\mathbf{r}-\mathbf{r}_E|^3}(\mathbf{r}-\mathbf{r}_E) - \frac{GM_M}{|\mathbf{r}-\mathbf{r}_M|^3}(\mathbf{r}-\mathbf{r}_M)$$

This system has no closed form analytical solution which is precisely why **numerical methods** such as Runge-Kutta (RK4) are essential to approximate the trajectory computationally.
Final answer:

Yes. ODEs derived from Newton's gravitational law model the spacecraft's trajectory as a systm of 6 first-order equations. For Artemis II, the Earth–Moon three-body interaction makes an analytical solution impossible, so numerical methods like RK4 are used to approximate the orbit.

2. Solve the following exercises: Exercise 3 from page 143 of [3].
Find $(\partial/\partial s)(f \circ T)(1, 0)$, where $f(u, v) = \cos u \sin v$ and $T(s, t) = (\cos(t^2 s), \log\sqrt{1+s^2})$.

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We apply the **Chain Rule** for composite functions. If $T(s,t) = (u(s,t),\, v(s,t))$, then:

$$\frac{\partial}{\partial s}(f \circ T) = \frac{\partial f}{\partial u}\frac{\partial u}{\partial s} + \frac{\partial f}{\partial v}\frac{\partial v}{\partial s}$$

Identify the components:

$$u(s,t) = \cos(t^2 s), \qquad v(s,t) = \log\sqrt{1+s^2} = \tfrac{1}{2}\log(1+s^2)$$

Partial derivatives of f:

$$\frac{\partial f}{\partial u} = -\sin u \sin v, \qquad \frac{\partial f}{\partial v} = \cos u \cos v$$

Partial derivatives of T with respect to s:

$$\frac{\partial u}{\partial s} = -t^2 \sin(t^2 s), \qquad \frac{\partial v}{\partial s} = \frac{s}{1+s^2}$$

Evaluate T at $(s,t) = (1,0)$:

$$u(1,0) = \cos(0) = 1, \qquad v(1,0) = \tfrac{1}{2}\log(2)$$

Evaluate the partial deivatives of T at $(1,0)$:

$$\left.\frac{\partial u}{\partial s}\right|_{(1,0)} = -0 \cdot \sin(0) = 0, \qquad \left.\frac{\partial v}{\partial s}\right|_{(1,0)} = \frac{1}{1+1} = \frac{1}{2}$$

Evaluate the partial derivatives of f at $(u,v) = (1,\, \frac{1}{2}\log 2)$:

$$\left.\frac{\partial f}{\partial u}\right|_{(1,\frac{1}{2}\log 2)} = -\sin(1)\sin\left(\tfrac{1}{2}\log 2\right), \qquad \left.\frac{\partial f}{\partial v}\right|_{(1,\frac{1}{2}\log 2)} = \cos(1)\cos\left(\tfrac{1}{2}\log 2\right)$$

Assemble the result:

$$\frac{\partial}{\partial s}(f \circ T)(1,0) = \left(-\sin 1 \sin\tfrac{\log 2}{2}\right)(0) + \left(\cos 1 \cos\tfrac{\log 2}{2}\right)\left(\frac{1}{2}\right)$$

$$= \frac{1}{2}\cos(1)\cos\left(\frac{\log 2}{2}\right)$$

Final answer:

$$\boxed{\frac{\partial}{\partial s}(f \circ T)(1,0) = \frac{1}{2}\cos(1)\cos\left(\frac{\log 2}{2}\right)}$$

3. Study and perform a writen analysis of the following examples: Example 6 and 8 from pages 153 and 154 of [3] respectively.

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Example 6: Gravitational Force as a Gradient Field
Setup and Formula

The gravitational force on a unit mass m at a point $(x, y, z) \in \mathbb{R}^3$, produced by a mass M located at the origin, is given by Newton's law:

$$\mathbf{F} = -\frac{GmM}{r^2}\mathbf{n}$$

where:

- G is the universal gravitational constant,
- $r = \|\mathbf{r}\| = \sqrt{x^2 + y^2 + z^2}$ is the distance from (x, y, z) to the origin,
- $\mathbf{n} = \mathbf{r}/r$ is the unit vector pointin from the origin toward (x, y, z),
- $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ is the position vector.

Key Result: F is a Conservative Field

The central claim is that $\mathbf{F}$ can be expressed as the negative gradient of the gravitational potential:

$$V = -\frac{GmM}{r}$$

That is:

$$\mathbf{F} = \nabla\left(\frac{GmM}{r}\right) = -\nabla V$$

This means $\mathbf{F}$ is a *conservative vector field* — a fundamental concept in physics and multivariable calculus. A conservative field does no net work over any closed path, and the work done between two points depends only on the endpoints, not on the path taken.

Geometric Interpretation

Since $V = -GmM/r$, the **level surfaces** of V are sets where $r = \text{const}$, i.e., **spheres centered at the origin**. By Theorem 14, the gradient ∇V is always perpendicular to the level surfaces of V. Therefore:

> $\mathbf{F}$ is **normal to every sphere centered at the origin**, pointing radially inward (toward M).

This is physically intuitive: gravity pulls mass m straight toward the source mass M, not tangentially. The force has no component along the sphere — it is purely radial.

Direction of the Force

Note that the negative sign in $\mathbf{F} = -\nabla V$ ensures the force points **inward** (toward decreasing r, i.e., toward the origin where M resides), consistent with the attractive nature of gravity.

Example 8: Electric Field and Equipotential Surfaces

Setup

Consider two electrical conductors: one with positive charge and one with negative charge. Between them, an **electric potential** is established, described by a scalar function:

$$\phi : \mathbb{R}^3 \to \mathbb{R}$$

The corresponding **electric field** is defined as:

$$\mathbf{E} = -\nabla\phi$$

Key Result: E is Perpendicular to Equipotential Surfaces

By Theorem 14, the gradient $\nabla\phi$ is perpendicular to the level surfaces of ϕ, i.e., to the surfaces where $\phi = \text{const}$. Since $\mathbf{E} = -\nabla\phi$, it follows that:

> **The electric field E is perpendicular (orthogonal) to every equipotential surface.**

These level surfaces $\{\phi = c\}$ are called *equipotential surfaces*, because the electric potential is constant on each of them — no work is done by the electric field when a charge moves along such a surface.

Geometric Interpretation (Figure 2.5.7)

The figure in the text illustrates this clearly:

- The **curved closed surfaces** represent equipotential surfaces ($\phi = \text{const}$).
- The **arrows labeled E** point from the region of higher potential to the region of lower potential, always crossing the equipotential lines **at right angles**.
- This orthogonality is a direct consequence of the gradient theorem.

Physical Meaning

This result has deep physical significance:

1. **No work along equipotentials:** Because $\mathbf{E} \perp$ equipotential surface, a test charge moved along such a surface experiences no force component in its direction of motion — hence no work is done, consistenith ϕ being constant there.

2. **Field lines and equipotentials are dual structures:** The field lines of $\mathbf{E}$ and the equipotential surfaces form an orthogonal network in space, visible in Figure 2.5.7.

3. **Direction of E:** The electric field points from high potential to low potential (the negative sign in $\mathbf{E} = -\nabla\phi$ ensures this), meaning positive charges are pushed from high to low potential.

Connecting Both Examples
Both examples are instances of the same fundamental principle:

Concept	**Example 6 (Gravity)**	**Example 8 (Electrostatics)**
Scalar potential	$V = -GmM/r$	$\phi : \mathbb{R}^3 \to \mathbb{R}$
Vector field	$\mathbf{F} = -\nabla V$	$\mathbf{E} = -\nabla\phi$
Level surfaces	Spheres ($r = \text{const}$)	Equipotential surfaces
Field direction	Perpendicular to spheres, inward	Perpendicular to equipotentials, high→low ϕ
Field type	Conservative	Conservative

Both $\mathbf{F}$ and $\mathbf{E}$ are **conservative fields** expressible as gradients of scalar potentials This guarantees path-independence of work and the orthogonality of field lines to level surfaces — a cornerstone result of vector calculus with far-reaching consequences in physics and engineering.

4. Read Galileo on page 163 of [3] and The Potential Equation on page 164.

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Space Travel & Orbital Mechanics: From Fourier to Artemis

The Mathematical Foundation
The PDF touches on something deeply relevant to modern space exploration: the universe speaks in differential equations. As the text notes, much of what we know about nature — from gravity to heat flow — can be expressed mathematically, and this is not a coincidence. Newton's law of gravitation and the calculus he developed allowed scientists to derive Kepler's three laws of planetary motion directly, which is exactly the mathematical backbone behind missions like Artemis.

The Heat Equation and Why It Matters for Space
The text introduces Fourier's heat equation:

$$k\left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2}\right) = \frac{\partial T}{\partial t}$$

This is not just an abstract formula, for a spacecaft like Artemis's Orion capsule, thermal management is critical — during lunar re-entry, the capsule faces extreme temperatures, engineers use exactly this type of partial differential equation to model how heat flows through the heat shield and into the capsule walls over

time. The function $T(x, y, z, t)$ describes tempeture at every point in the material at every moment, and solving it numerically is what keeps astronauts alive.

The Potential Equation (Laplace's Equation)
The second major equation in the reading is Laplace's equation:

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0$$

Here, $V = -GmM/r$ is the gravitational potential produced by a mass M at distance r. This satisfies Laplace's equation everywhere except at the source mass itself. If you know the gravitational potential field around the Moon and Earth, you can determine the force on the spacecraft at every point in space via $\mathbf{F} = -\nabla V$. This is exactly how trajectory computers for Artemis work — they compute the potential field and take its gradient, and as the mission enters the gravitational influence of the Moon, the potential function shifts and the trajectory curves accordingly.

From PDEs to ODEs: Orbital Trajectories
The gravitational force law leads to a system of second-order ODEs for the spacecraft's position $\mathbf{r}(t) = (x(t), y(t), z(t))$:

$$\frac{d^2\mathbf{r}}{dt^2} = -\frac{GM}{|\mathbf{r}|^3}\mathbf{r}$$

This is Newton's second law applied to the gravitational force. For Artemis, this must be solved for a three-body system (Earth + Moon + spacecraft), which has no clean analytical solution — numerical methods are essential. Methods like Runge-Kutta are usd to integrate these equations step by step through time, giving of the spacecraft position and velocity at each moment of the mission.

The Big Picture
What the reading illustrates nd what this course builds toward — is that: the gravitational potential V satisfies Laplace's equation and gives the force field; Newton's second law turns that force into a system of ODEs for position over time; numerical methods solve those ODEs when analytical solutions do not exist andthermal PDEs run in parallel to keep the engineering safe. Missions like Artemis are, at their mathematical core, an enormous numerical ODE/PDE slving problem. The elegance is that all of it traces back to the equations in those two pages — Fourier's nineteenth-century work on heat and Laplace's gravitational potential are still alive inside every space mission today.

5. Solve exercise 19 from page 167 of [3].

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Show that the Newtonian potential V (see Example 6, Section 2.5) satisfies Laplace's equation

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0 \quad \text{for} \quad (x, y, z) \neq (0, 0, 0).$$

EXAMPLE 6 - Section 2.5 The gravitational force on a unit mass m at (x, y, z) produced by a mass M at the origin in $\mathbb{R}^3$, according to Newton's law of gravitation, is given by

$$\mathbf{F} = -\frac{GmM}{r^2}\,\mathbf{n},$$

where G is a constant; $r = \|\mathbf{r}\| = \sqrt{x^2 + y^2 + z^2}$ is the distance from (x, y, z) to the origin; and $\mathbf{n} = \mathbf{r}/r$, the unit vector in the direction of $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, which is the position vector from the origin to (x, y, z).
Note that $\mathbf{F} = \nabla\big(\frac{GmM}{r}\big) = -\nabla V$, that is, $\mathbf{F}$ is the negative of the gradient of the gravitational potential $V = -\frac{GmM}{r}$. This can be verified as in Example 1. Note that $\mathbf{F}$ is directed inward, toward the origin. Moreover, the level surfaces of V are spheres. $\mathbf{F}$ is normal to these spheres, which confirms the result of Theorem 14.

Process:
The Newtonian potential is given by

$$V = -\frac{GmM}{r}, \qquad r = \sqrt{x^2 + y^2 + z^2},$$

where GmM is a nonzero constant. Since constant factors pass through differentiation, it suffices to show that $u = \frac{1}{r}$ satisfies Laplace's equation for $(x, y, z) \neq (0, 0, 0)$.

First partial derivatives of r. For any coordinate, say x:

$$\frac{\partial r}{\partial x} = \frac{x}{r}.$$

First partial derivative of $u = 1/r$.

$$\frac{\partial u}{\partial x} = -\frac{1}{r^2}\frac{\partial r}{\partial x} = -\frac{x}{r^3}.$$

Second partial derivative of u with respect to x.

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x}\Big(-\frac{x}{r^3}\Big) = -\frac{r^3 - x \cdot 3r^2\dfrac{\partial r}{\partial x}}{r^6} = -\frac{r^3 - 3x^2 r}{r^6} = \frac{3x^2 - r^2}{r^5}.$$

By symmetry in x, y, z:

$$\frac{\partial^2 u}{\partial y^2} = \frac{3y^2 - r^2}{r^5}, \qquad \frac{\partial^2 u}{\partial z^2} = \frac{3z^2 - r^2}{r^5}.$$

Summing the three terms:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = \frac{3(x^2 + y^2 + z^2) - 3r^2}{r^5} = \frac{3r^2 - 3r^2}{r^5} = 0,$$

where we used $x^2 + y^2 + z^2 = r^2$. Since $V = -GmM \cdot u$, it follows immediately that

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = -GmM\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}\right) = 0.$$

Final answer:

For $(x, y, z) \neq (0, 0, 0)$, writing $r = \sqrt{x^2 + y^2 + z^2}$ and $u = 1/r$, one computes $\partial^2 u/\partial x^2 = (3x^2 - r^2)/r^5$ and its cyclic analogues. Summing gives

$$\Delta u = \frac{3(x^2 + y^2 + z^2) - 3r^2}{r^5} = 0,$$

hence $\Delta V = -GmM\,\Delta u = 0$, so V satisfies Laplace's equation.

1. Revisar y presentar un análisis escrito del ejemplo 7 de la página 197 en [3].

P15.6 — David

2. Revisar y presentar un análisis escrito del ejemplo 9 la página 200 en [3].

P15.7 — David

3. Revisar y presentar un análisis escrito del Ejemplo 14 de la página 201 en [3].

P15.8 — David

4. Revisar y presentar un análisis escrito del Ejemplo 3 de la página 213 en [3].

P15.9 — David

5. Resolver el ejercicio 1 de la página 219 en [3].

P15.10 — David

6. Leer y hacer un resumen de la Sección 4.5 de la página 291 de [3].

P15.11 — David

7. Revisar y hacer un análisis del Ejemplo 7 en la página 396 de [3].

P15.12 — David

Routine Problems:

It is expected that the following activities will allow you to understand, review, and use analytical methods: integrating factor, separable variables, and exact equations (among others). Furthermore, that the development of these activities generates feelings of confidence in you when solving some first-order ordinary differential equations.

Activity 1. Perform exercises 1 through 16 on page 33 of [2], submit 4, 8, 10, 12, 13, 14, and 16.

In each of Problems 1 through 8:

1. Draw a direction field for the given differential equation.
2. Based on an inspection of the direction field, describe how solutions behave for large t.
3. Find the general solution of the given differential equation, and use it to determine how solutions behave as $t \to \infty$.

---

4. $y' + \frac{1}{t}y = 3\cos(2t), \quad t > 0$

Desarrollo y Respuesta

1.Draw a direction field for the given differential equation.

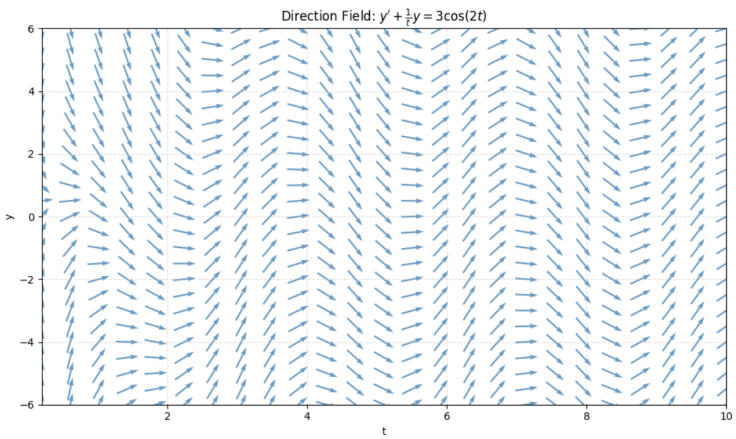


Figura 2: Direction field for $y' + \frac{1}{t}y = 3\cos(2t)$

2.Based on an inspection of the direction field, describe how solutions behave for large t.
For large t, the direction field shows that all solutions appear to oscillate in a wave-like pattern. Regardless of the initial condition, the solutions seem to be attracted toward the same repeating oscillation, suggesting that any transient behavior dies out over time and every solution eventually behaves like a sinusoidal wave of fixed amplitude.
3.Find the general solution of the given differential equation, and use it to determine how solutions behave as $t \to \infty$.

The equation is already in standard form $y' + \frac{1}{t}y = 3\cos(2t)$, so the integrating factor is:

$$\mu(t) = e^{\int \frac{1}{t}\,dt} = e^{\ln t} = t$$

Multiplying both sides by $\mu(t) = t$:

$$ty' + y = 3t\cos(2t) \implies \frac{d}{dt}(ty) = 3t\cos(2t)$$

Integrating both sides (by parts on the right side, with $u = 3t$, $dv = \cos(2t)\,dt$):

$$ty = \frac{3t}{2}\sin(2t) + \frac{3}{4}\cos(2t) + C$$

Dividing by t, the general solution is:

$$y = \frac{3}{2}\sin(2t) + \frac{3}{4t}\cos(2t) + \frac{C}{t}$$

As $t \to \infty$, the transient terms vanish since $\frac{3}{4t}\cos(2t) \to 0$ and $\frac{C}{t} \to 0$, so all solutions satisfy:

$$y \to \frac{3}{2}\sin(2t) \quad \text{as } t \to \infty$$

regardless of the initial condition.

---

8. $2y' + y = 3t^2$

Desarrollo y Respuesta

1.Draw a direction field for the given differential equation.

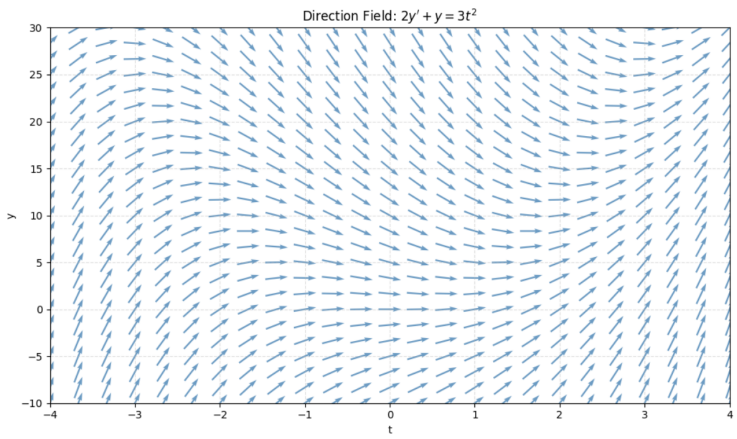


Figura 3: Direction field for $2y' + y = 3t^2$

2.Based on an inspection of the direction field, describe how solutions behave for large t.

For large t, the direction field shows that all solutions grow without bound. The slopes become increasingly steep and positive as t increases, suggesting that every solution diverges to infinity regardless of the initial condition.

3.Find the general solution of the given differential equation, and use it to determine how solutions behave as $t \to \infty$.

Rewriting in standard form by dividing by 2:

$$y' + \frac{1}{2}y = \frac{3}{2}t^2$$

The integrating factor is:

$$\mu(t) = e^{\int \frac{1}{2}\,dt} = e^{t/2}$$

Multiplying both sides by $\mu(t) = e^{t/2}$:

$$\frac{d}{dt}\left(e^{t/2}y\right) = \frac{3}{2}t^2e^{t/2}$$

Integrating the right side by parts twice:

$$e^{t/2}y = 3t^2e^{t/2} - 12te^{t/2} + 24e^{t/2} + C$$

Dividing by $e^{t/2}$, the general solution is:

$$y = 3t^2 - 12t + 24 + Ce^{-t/2}$$

As $t \to \infty$, the transient term $Ce^{-t/2} \to 0$, so all solutions behave like:

$$y \to 3t^2 - 12t + 24 \quad \text{as } t \to \infty$$

which grows without bound, confirming that all solutions diverge to $+\infty$ regardless of the initial condition.

In each of Problems 9 through 12, find the solution of the given initial value problem.

---

10. $y' + 2y = te^{-2t}, \quad y(1) = 0$

Desarrollo y Respuesta

Integrating Factor. This is a first-order linear ODE of the form $y' + p(t)y = g(t)$, with $p(t) = 2$. The integrating factor is:

$$\mu(t) = e^{\int 2\,dt} = e^{2t}$$

Multiplying both sides by $\mu(t) = e^{2t}$:

$$e^{2t}y' + 2e^{2t}y = te^{-2t} \cdot e^{2t} = t$$

The left-hand side is an exact derivative:

$$\frac{d}{dt}\left(e^{2t}y\right) = t$$

Integrating both sides with respect to t:

$$e^{2t}y = \int t\,dt = \frac{t^2}{2} + C$$

General solution:

$$y = \left(\frac{t^2}{2} + C\right)e^{-2t}$$

Applying the initial condition $y(1) = 0$:

$$0 = \left(\frac{1}{2} + C\right)e^{-2} \implies \frac{1}{2} + C = 0 \implies C = -\frac{1}{2}$$

Solution:

$$\boxed{y = \frac{t^2 - 1}{2}\,e^{-2t}}$$

---

12. $ty' + (t+1)y = t, \quad y(\ln 2) = 1, \quad t > 0$

Desarrollo y Respuesta

Standard Form. Divide both sides by t to write the equation in standard linear form:

$$y' + \frac{t+1}{t}\,y = 1 \quad \implies \quad y' + \left(1 + \frac{1}{t}\right)y = 1$$

Integrating Factor. With $p(t) = 1 + \frac{1}{t}$:

$$\mu(t) = e^{\int\left(1+\frac{1}{t}\right)dt} = e^{t+\ln t} = te^t$$

Multiplying both sides by $\mu(t) = te^t$:

$$te^t y' + (t+1)e^t y = te^t$$

The left-hand side is an exact derivative:

$$\frac{d}{dt}\left(te^t y\right) = te^t$$

Integrating both sides with respect to t (using integration by parts on the right):

$$te^t y = \int te^t\,dt = e^t(t-1) + C$$

General solution:

$$y = \frac{t-1}{t} + \frac{Ce^{-t}}{t}$$

Applying the initial condition $y(\ln 2) = 1$:

$$1 = \frac{\ln 2 - 1}{\ln 2} + \frac{Ce^{-\ln 2}}{\ln 2} = \frac{\ln 2 - 1}{\ln 2} + \frac{C}{2\ln 2}$$

$$1 - \frac{\ln 2 - 1}{\ln 2} = \frac{C}{2\ln 2} \implies \frac{1}{\ln 2} = \frac{C}{2\ln 2} \implies C = 2$$

Solution:

$$\boxed{y = \frac{t - 1 + 2e^{-t}}{t}}$$

In each of Problems 13 and 14:

1. Draw a direction field for the given differential equation. How do solutions appear to behave as t becomes large? Does the behavior depend on the choice of the initial value a? Let a_0 be the value of a for which the transition from one type of behavior to another occurs. Estimate the value of a_0.
2. Solve the initial value problem and find the critical value a_0 exactly.
3. Describe the behavior of the solution corresponding to the initial value a_0.

---

13. $y' - \frac{1}{2}y = 2\cos(t), \quad y(0) = a$

Desarrollo y Respuesta

1. Draw a direction field for the given differential equation. How do solutions appear to behave as t becomes large? Does the behavior depend on the choice of the initial value a? Let a_0 be the value of a for which the transition from one type of behavior to another occurs. Estimate the value of a_0.

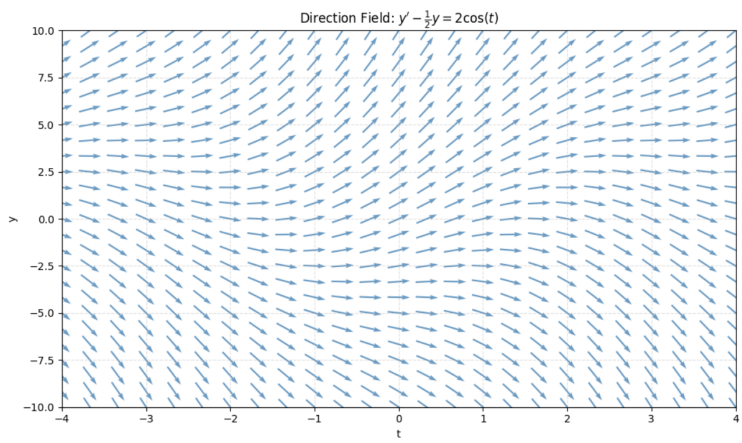


Figura 4: Direction field for $y' - \frac{1}{2}y = 2\cos(t)$

The solutions clearly diverge between $-\infty$ and $+\infty$, and the long-term behavior depends entirely on the choice of the initial value a. Observing the direction field, there exists a critical value a_0 at which the transition between these two types of behavior occurs. For initial values $a > a_0$ the solutions grow toward $+\infty$, while for $a < a_0$ they tend toward $-\infty$. From the direction field, this critical value can be estimated as $a_0 \approx -1{,}3$.

2. Solve the initial value problem and find the critical value a_0 exactly.

We solve the equation $y' - \frac{1}{2}y = 2\cos(t)$ using the integrating factor method.

The integrating factor is:

$$\mu(t) = e^{\int -\frac{1}{2}\,dt} = e^{-\frac{t}{2}}$$

Multiplying both sides by $\mu(t)$:

$$\frac{d}{dt}\left(e^{-\frac{t}{2}}\,y\right) = 2e^{-\frac{t}{2}}\cos(t)$$

Integrating the right-hand side using the formula $\displaystyle\int e^{at}\cos(bt)\,dt = \frac{e^{at}(a\cos(bt) + b\sin(bt))}{a^2+b^2}$, with $a = -\frac{1}{2}$ and $b = 1$:

$$\int 2e^{-\frac{t}{2}}\cos(t)\,dt$$

$$= 2 \cdot \frac{e^{-\frac{t}{2}}\left(-\frac{1}{2}\cos(t) + \sin(t)\right)}{\frac{1}{4}+1} + C$$

$$= \frac{8}{5}e^{-\frac{t}{2}}\left(-\frac{1}{2}\cos(t) + \sin(t)\right) + C$$

Thus:

$$e^{-\frac{t}{2}}\,y = \frac{8}{5}e^{-\frac{t}{2}}\left(-\frac{1}{2}\cos(t) + \sin(t)\right) + C$$

Multiplying through by $e^{\frac{t}{2}}$:

$$y(t) = \frac{8}{5}\left(-\frac{1}{2}\cos(t) + \sin(t)\right) + Ce^{\frac{t}{2}} = -\frac{4}{5}\cos(t) + \frac{8}{5}\sin(t) + Ce^{\frac{t}{2}}$$

Applying the initial condition $y(0) = a$:

$$a = -\frac{4}{5}\cos(0) + \frac{8}{5}\sin(0) + C = -\frac{4}{5} + C \implies C = a + \frac{4}{5}$$

So the solution to the IV is:

$$\boxed{y(t) = -\frac{4}{5}\cos(t) + \frac{8}{5}\sin(t) + \left(a + \frac{4}{5}\right)e^{\frac{t}{2}}}$$

Since $e^{t/2} \to +\infty$ as $t \to \infty$, the long-term behavior is governed entirely by the sign of the coefficient $C = a + \frac{4}{5}$:

- If $a > -\frac{4}{5}$, then $C > 0$ and $y(t) \to +\infty$.
- If $a < -\frac{4}{5}$, then $C < 0$ and $y(t) \to -\infty$.
- If $a = -\frac{4}{5}$, then $C = 0$ and $y(t)$ oscillates boundedly.

Therefore, the exact critical value is:

$$a_0 = -\frac{4}{5}$$

This confirms and refines the estimate $a_0 \approx -1{,}3$ from the direction field (the true value $-0{,}8$ lies in the same region).

3. Describe the behavior of the solution corresponding to the initial value a_0.

When $a = a_0 = -\frac{4}{5}$, the constant $C = 0$, and the solution reduces to:

$$y(t) = -\frac{4}{5}\cos(t) + \frac{8}{5}\sin(t)$$

This can be written in amplitude-phase form as $y(t) = R\cos(t - \phi)$, where:

$$R = \sqrt{\left(\frac{4}{5}\right)^2 + \left(\frac{8}{5}\right)^2} = \sqrt{\frac{16+64}{25}} = \sqrt{\frac{80}{25}} = \frac{4\sqrt{5}}{5}$$

Therefore, when $a = a_0$, the solution is purely oscillatory with amplitude $\frac{4\sqrt{5}}{5} \approx 1{,}789$ and period 2π. It neither grows nor decays as $t \to \infty$, in contrast to all other solutions which are eventually dominated by the $e^{t/2}$ term and diverge to $\pm\infty$.

---

14. $3y' - 2y = e^{-\pi t/2}, \quad y(0) = a$

Desarrollo y Respuesta

1. Draw a direction field for the given differential equation. How do solutions appear to behave as t becomes large? Does the behavior depend on the choice of the initial value a? Let a_0 be the value of a for which the transition from one type of behavior to another occurs. Estimate the value of a_0.

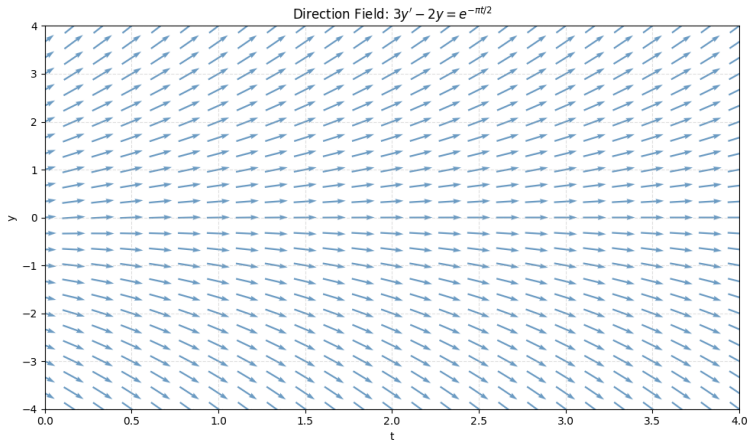


Figura 5: Direction field for $3y' - 2y = e^{-\pi t/2}, \quad y(0) = a$

The solutions clearly diverge between $-\infty$ and $+\infty$, and the long-term behavior depends entirely on the choice of the initial value a. Observing the direction field, there exists a critical value a_0 at which the transition between these two types of behavior occurs. For initial values $a > a_0$ the solutions grow toward $+\infty$, while for $a < a_0$ they tend toward $-\infty$. From the direction field, this critical value can be estimated as $a_0 \approx 0$. Even when this is an approximation I'm almost completely sure that is the critical value.

2. Solve the initial value problem and find the critical value a_0 exactly. We rewrite the equation in standard form:

$$y' - \frac{2}{3}y = \frac{1}{3}e^{-\pi t/2}$$

The integrating factor is:

$$\mu(t) = e^{\int -\frac{2}{3}\,dt} = e^{-\frac{2}{3}t}$$

Multiplying both sides by $\mu(t)$:

$$\frac{d}{dt}\left[e^{-\frac{2}{3}t}\,y\right] = \frac{1}{3}e^{-\frac{2}{3}t} \cdot e^{-\frac{\pi}{2}t} = \frac{1}{3}e^{-\left(\frac{2}{3}+\frac{\pi}{2}\right)t}$$

Integrating both sides:

$$e^{-\frac{2}{3}t}\,y = \frac{1}{3} \cdot \frac{1}{-\left(\frac{2}{3}+\frac{\pi}{2}\right)}e^{-\left(\frac{2}{3}+\frac{\pi}{2}\right)t} + C$$

$$e^{-\frac{2}{3}t}\,y = \frac{-1}{3\left(\frac{2}{3}+\frac{\pi}{2}\right)}e^{-\left(\frac{2}{3}+\frac{\pi}{2}\right)t} + C$$

Simplifying the constant. Let $k = \dfrac{2}{3} + \dfrac{\pi}{2} = \dfrac{4+3\pi}{6}$, so:

$$e^{-\frac{2}{3}t}\,y = \frac{-1}{3k}\,e^{-kt} + C$$

Multiplying both sides by $e^{\frac{2}{3}t}$:

$$y = \frac{-1}{3k}\, e^{-\left(k-\frac{2}{3}\right)t} + Ce^{\frac{2}{3}t}$$

Since $k - \dfrac{2}{3} = \dfrac{\pi}{2}$, we get:

$$y = \frac{-1}{3k}\, e^{-\frac{\pi}{2}t} + Ce^{\frac{2}{3}t}$$

Applying the initial condition $y(0) = a$:

$$a = \frac{-1}{3k} + C \implies C = a + \frac{1}{3k}$$

So the solution is:

$$\boxed{y(t) = \frac{-1}{3k}\, e^{-\frac{\pi}{2}t} + \left(a + \frac{1}{3k}\right) e^{\frac{2}{3}t}}, \qquad k = \frac{4+3\pi}{6}$$

Finding the critical value a_0:
As $t \to \infty$, the term $e^{-\frac{\pi}{2}t} \to 0$, so the long-term behavior is governed entirely by the term $\left(a + \dfrac{1}{3k}\right) e^{\frac{2}{3}t}$.

- If $a + \dfrac{1}{3k} > 0$, then $y \to +\infty$.
- If $a + \dfrac{1}{3k} < 0$, then $y \to -\infty$.
- If $a + \dfrac{1}{3k} = 0$, the exponential growth term vanishes and $y \to 0$.

Therefore, the exact critical value is:

$$\boxed{a_0 = -\frac{1}{3k} = -\frac{2}{4+3\pi} \approx -0{,}132}$$

This confirms our earlier estimate: a_0 is close to 0, but the exact value is $a_0 = -\dfrac{2}{4+3\pi}$.

3. Describe the behavior of the solution corresponding to the initial value a_0.

For the critical initial value $a_0 = -\dfrac{2}{4+3\pi}$, the coefficient of the growing exponential vanishes, i.e., $a_0 + \dfrac{1}{3k} = 0$. The solution reduces to:

$$y(t) = \frac{-1}{3k}\, e^{-\frac{\pi}{2}t}$$

Since $-\dfrac{1}{3k} < 0$ and the exponential $e^{-\frac{\pi}{2}t} \to 0$ as $t \to \infty$, this solution decays monotonically to zero from below. That is:

$$\lim_{t\to\infty} y(t) = 0$$

This is the unique solution that neither grows to $+\infty$ nor diverges to $-\infty$. It represents the boundary between the two divergent behaviors.

In each of Problems 15 and 16:

1. Draw a direction field for the given differential equation. How do solutions appear to behave as $t \to 0$? Does the behavior depend on the choice of the initial value a? Let a_0 be the critical value of a, that is, the initial value such that the solutions for $a < a_0$ and the solutions for $a > a_0$ have different behaviors as $t \to \infty$. Estimate the value of a_0.
2. Solve the initial value problem and find the critical value a_0 exactly.
3. Describe the behavior of the solution corresponding to the initial value a_0.

---

16. $(\sin(t))y' + (\cos(t))y = e^t, \quad y(1) = a, \quad 0 < t < \pi$

Desarrollo y Respuesta

1. Draw a direction field for the given differential equation. How do solutions appear to behave as $t \to 0$? Does the behavior depend on the choice of the initial value a? Let a_0 be the critical value of a, that is, the initial value such that the solutions for $a < a_0$ and the solutions for $a > a_0$ have different behaviors as $t \to \infty$. Estimate the value of a_0.

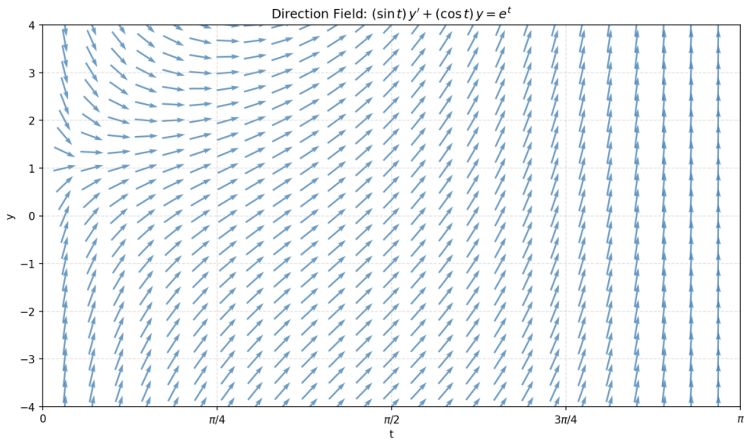


Figura 6: Direction field for $(\sin(t))y' + (\cos(t))y = e^t, \quad y(1) = a, \quad 0 < t < \pi$

From the direction field, as $t \to 0^+$ the slopes become increasingly steep, suggesting that solutions grow without bound near the left endpoint regardless of the initial value a; that is, this singular behavior does not depend on the choice of a. The D.F doesn't show any critical value in the long therm solutions of the possible functions, in that order of ideas when t tends to ∞ the solutions is as well up without any bound.

2. Solve the initial value problem and find the critical value a_0 exactly.

The left-hand side is an exact derivative:

$$(\sin t)\, y' + (\cos t)\, y = \frac{d}{dt}\big[\sin(t) \cdot y\big]$$

so the equation becomes:

$$\frac{d}{dt}\big[\sin(t) \cdot y\big] = e^t$$

Integrating both sides directly:

$$\sin(t) \cdot y = e^t + C$$

$$y = \frac{e^t + C}{\sin t}$$

Applying the initial condition $y(1) = a$:

$$a = \frac{e^1 + C}{\sin 1} \implies C = a\sin(1) - e$$

So the solution to the initial value problem is:

$$\boxed{y(t) = \frac{e^t + a\sin(1) - e}{\sin t}}$$

Finding the critical value a_0: As $t \to \pi^-$, we have $\sin t \to 0^+$, so the behavior of $y(t)$ is governed entirely by the sign of the numerator $e^t + a\sin(1) - e$ evaluated at $t = \pi$:

- If $e^\pi + a\sin(1) - e > 0$, then $y \to +\infty$.
- If $e^\pi + a\sin(1) - e < 0$, then $y \to -\infty$.
- If $e^\pi + a\sin(1) - e = 0$, the numerator vanishes and the transition occurs.

Setting the numerator equal to zero:

$$e^\pi + a_0\sin(1) - e = 0 \implies a_0 = \frac{e - e^\pi}{\sin(1)}$$

$$\boxed{a_0 = \frac{e - e^\pi}{\sin(1)} \approx -0{,}172}$$

3. Describe the behavior of the solution corresponding to the initial value a_0.

For the critical initial value $a_0 = \dfrac{e - e^\pi}{\sin(1)}$, the numerator of the solution vanishes as $t \to \pi^-$, so the growing exponential term is exactly cancelled. The solution reduces to:

$$y(t) = \frac{e^t - e^\pi}{\sin t}$$

which remains bounded and decreases toward zero as $t \to \pi^-$, since both numerator and denominator vanish at $t = \pi$. This solution represents the unique boundary case that neither diverges to $+\infty$ nor to $-\infty$, separating the two divergent behaviors observed for all other initial values.

Actividad 2. Realizar los ejercicios del 1 al 16 de la página 40 de [2], entregar 8, 10, 12, 14 y 16.

Ejercicio 8 (pág. 40, [2])

Enunciado

[Transcribir el enunciado del ejercicio 8 aquí.]

Desarrollo y Respuesta

Ejercicio 10 (pág. 40, [2])

Enunciado

[Transcribir el enunciado del ejercicio 10 aquí.]

Desarrollo y Respuesta

Ejercicio 12 (pág. 40, [2])

Enunciado

[Transcribir el enunciado del ejercicio 12 aquí.]

Desarrollo y Respuesta

Ejercicio 14 (pág. 40, [2])

Enunciado

[Transcribir el enunciado del ejercicio 14 aquí.]

Desarrollo y Respuesta

Ejercicio 16 (pág. 40, [2])

Enunciado

[Transcribir el enunciado del ejercicio 16 aquí.]

Desarrollo y Respuesta

Actividad 3. Realizar los ejercicios del 1 al 22 de la página 77 de [2], entregar 10, 12, 14, 16, 18 y 22.

Ejercicio 10 (pág. 77, [2])

Enunciado

Resolver el siguiente problema con valor inicial y determinar al menos aproximadamente dónde la solución es válida:

$$(9x^2 + y - 1) - (4y - x)y' = 0, \quad y(1) = 0$$

Desarrollo y Respuesta

Paso 1: Identificar el tipo de ecuación.
La ecuación tiene la forma $M(x, y) + N(x, y)y' = 0$, donde:

$$M = 9x^2 + y - 1 \qquad N = -(4y - x) = x - 4y$$

Para verificar si es exacta, calculamos las derivadas parciales cruzadas:

$$\frac{\partial M}{\partial y} = 1 \qquad \frac{\partial N}{\partial x} = 1$$

Como $\dfrac{\partial M}{\partial y} = \dfrac{\partial N}{\partial x}$, la ecuación **es exacta**.

Paso 2: Encontrar la función $F(x, y)$ tal que $F_x = M$ y $F_y = N$.
Integramos M respecto a x:

$$F = \int (9x^2 + y - 1)\, dx = 3x^3 + xy - x + h(y)$$

donde $h(y)$ es una función desconocida que solo depende de y, ya que al derivar respecto a x cualquier función pura de y desaparece.

Paso 3: Determinar $h(y)$ usando la segunda condición $F_y = N$.
Derivamos F respecto a y e igualamos a N:

$$\frac{\partial F}{\partial y} = x + h'(y) = x - 4y$$

Despejando:

$$h'(y) = -4y \implies h(y) = -2y^2$$

Paso 4: Escribir la solución general.
Juntando todo:

$$F(x, y) = 3x^3 + xy - x - 2y^2 = C$$

Paso 5: Aplicar la condición inicial $y(1) = 0$.
Sustituimos $x = 1$, $y = 0$:

$$3(1)^3 + (1)(0) - (1) - 2(0)^2 = C \implies C = 2$$

La solución del problema con valor inicial es:

$$\boxed{3x^3 + xy - x - 2y^2 = 2}$$

Paso 6: Validez de la solución.
Como M y N son polinomios, son continuas en todo $\mathbb{R}^2$. Por el Teorema de Existencia y Unicidad, la solución es válida en algún intervalo que contiene $x = 1$. Dado que no hay singularidades, la solución es válida para todo $x \in \mathbb{R}$, aunque la relación implícita puede no definir y como función explícita en todo ese dominio.

Ejercicio 12 (pág. 77, [2])

Enunciado

Encontrar el valor de b para el cual la siguiente ecuación es exacta, y luego resolverla:

$$\left(ye^{2xy} + x\right) + bxe^{2xy}y' = 0$$

Desarrollo y Respuesta

Paso 1: Identificar M y N.
La ecuación tiene la forma $M + Ny' = 0$, donde:

$$M = ye^{2xy} + x \qquad N = bxe^{2xy}$$

Paso 2: Encontrar b usando la condición de exactitud.
Para que la ecuación sea exacta se debe cumplir $\dfrac{\partial M}{\partial y} = \dfrac{\partial N}{\partial x}$.
Calculamos cada parcial:

$$\frac{\partial M}{\partial y} = e^{2xy} + y \cdot 2xe^{2xy} = e^{2xy}(1 + 2xy)$$

$$\frac{\partial N}{\partial x} = be^{2xy} + bx \cdot 2ye^{2xy} = be^{2xy}(1 + 2xy)$$

Igualando:

$$e^{2xy}(1 + 2xy) = be^{2xy}(1 + 2xy) \implies \boxed{b = 1}$$

Paso 3: Reescribir la ecuación con $b = 1$.

$$\left(ye^{2xy} + x\right) + xe^{2xy}y' = 0$$

Paso 4: Encontrar $F(x, y)$ tal que $F_x = M$ y $F_y = N$.
Integramos $N = xe^{2xy}$ respecto a y:

$$F = \int xe^{2xy}\, dy = \frac{1}{2}e^{2xy} + g(x)$$

donde $g(x)$ es una función desconocida que solo depende de x.

Paso 5: Determinar $g(x)$ usando la condición $F_x = M$.
Derivamos F respecto a x e igualamos a M:

$$\frac{\partial F}{\partial x} = ye^{2xy} + g'(x) = ye^{2xy} + x$$

Despejando:

$$g'(x) = x \implies g(x) = \frac{x^2}{2}$$

Paso 6: Escribir la solución general.
Juntando todo:

$$F(x, y) = \frac{1}{2}e^{2xy} + \frac{x^2}{2} = C$$

Multiplicando por 2:

$$\boxed{e^{2xy} + x^2 = C}$$

Ejercicio 14 (pág. 77, [2])

Enunciado

Mostrar que cualquier ecuación separable de la forma

$$M(x) + N(y)\, y' = 0$$

es también una ecuación exacta.

Desarrollo y Respuesta

Planteamiento.
Una ecuación de la forma $M(x) + N(y)y' = 0$ es separable porque M depende *solo* de x y N depende *solo* de y.

Verificación de la condición de exactitud.
Para que sea exacta se debe cumplir:

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Como M depende **solo de** x:

$$\frac{\partial M}{\partial y} = 0$$

Como N depende **solo de** y:

$$\frac{\partial N}{\partial x} = 0$$

Por lo tanto:

$$\frac{\partial M}{\partial y} = 0 = \frac{\partial N}{\partial x} \checkmark$$

La condición de exactitud se cumple para **cualquier** $M(x)$ y $N(y)$, independientemente de su forma. Luego, toda ecuación separable de esta forma es también exacta. ■

Nota. La verificación no depende de los valores específicos de M o N, sino únicamente de que cada función dependa de una sola variable. Eso garantiza que sus derivadas parciales respecto a la otra variable sean siempre cero.

Ejercicio 16 (pág. 77, [2])

Enunciado

Mostrar que la siguiente ecuación no es exacta, pero que se vuelve exacta al multiplicarla por el factor integrante dado. Luego resolver la ecuación:

$$(x+2)\sin(y) + (x\cos(y))\,y' = 0, \qquad \mu(x,y) = xe^x$$

Desarrollo y Respuesta

Paso 1: Verificar que la ecuación original NO es exacta.
Identificamos $M = (x+2)\sin(y)$ y $N = x\cos(y)$:

$$\frac{\partial M}{\partial y} = (x+2)\cos(y) \qquad \frac{\partial N}{\partial x} = \cos(y)$$

Como $(x+2)\cos(y) \neq \cos(y)$, la ecuación **no es exacta**.

Paso 2: Multiplicar por el factor integrante $\mu = xe^x$.

$$xe^x(x+2)\sin(y) + x^2e^x\cos(y)\,y' = 0$$

Ahora: $M^* = xe^x(x+2)\sin(y)$ y $N^* = x^2e^x\cos(y)$.

Paso 3: Verificar exactitud de la ecuación modificada.

$$\frac{\partial M^*}{\partial y} = xe^x(x+2)\cos(y)$$

$$\frac{\partial N^*}{\partial x} = (2xe^x + x^2e^x)\cos(y) = xe^x(x+2)\cos(y)$$

Como $\dfrac{\partial M^*}{\partial y} = \dfrac{\partial N^*}{\partial x}$, la ecuación modificada **es exacta**. ✓

Paso 4: Encontrar $F(x,y)$ tal que $F_x = M^*$ y $F_y = N^*$.
Integramos N^* respecto a y:

$$F = \int x^2e^x\cos(y)\,dy = x^2e^x\sin(y) + g(x)$$

Paso 5: Determinar $g(x)$ usando $F_x = M^*$.

$$\frac{\partial F}{\partial x} = (2xe^x + x^2e^x)\sin(y) + g'(x) = xe^x(x+2)\sin(y) + g'(x)$$

Igualando a M^*:

$$g'(x) = 0 \implies g(x) = 0$$

Paso 6: Solución general.

$$\boxed{x^2e^x\sin(y) = C}$$

Verificación. Derivando implícitamente respecto a x:

$$(2xe^x + x^2e^x)\sin(y) + x^2e^x\cos(y)\,y' = 0$$

$$xe^x(x+2)\sin(y) + x^2e^x\cos(y)\,y' = 0$$

Dividiendo entre xe^x:

$$(x+2)\sin(y) + x\cos(y)\,y' = 0\checkmark$$

Ejercicio 18 (pág. 77, [2])

Enunciado

Encontrar un factor integrante y resolver la ecuación:

$$(3x^2y + 2xy + y^3) + (x^2 + y^2)\,y' = 0$$

Desarrollo y Respuesta

Paso 1: Verificar que la ecuación NO es exacta.
Identificamos $M = 3x^2y + 2xy + y^3$ y $N = x^2 + y^2$:

$$\frac{\partial M}{\partial y} = 3x^2 + 2x + 3y^2 \qquad \frac{\partial N}{\partial x} = 2x$$

Como $3x^2 + 2x + 3y^2 \neq 2x$, la ecuación **no es exacta**.

Paso 2: Encontrar el factor integrante.
Calculamos el cociente $\dfrac{M_y - N_x}{N}$:

$$\frac{M_y - N_x}{N} = \frac{(3x^2 + 2x + 3y^2) - 2x}{x^2 + y^2} = \frac{3(x^2 + y^2)}{x^2 + y^2} = 3$$

Como el resultado depende **solo de** x, el factor integrante es:

$$\mu(x) = e^{\int 3\,dx} = e^{3x}$$

Paso 3: Multiplicar por $\mu = e^{3x}$ y verificar exactitud.

$$M^* = e^{3x}(3x^2y + 2xy + y^3) \qquad N^* = e^{3x}(x^2 + y^2)$$

$$\frac{\partial M^*}{\partial y} = e^{3x}(3x^2 + 2x + 3y^2)$$

$$\frac{\partial N^*}{\partial x} = 3e^{3x}(x^2 + y^2) + e^{3x}(2x) = e^{3x}(3x^2 + 3y^2 + 2x)\checkmark$$

La ecuación modificada **es exacta**.

Paso 4: Encontrar $F(x,y)$ integrando $F_x = M^*$ respecto a x.
Dado que $M^* = e^{3x}(3x^2y + 2xy + y^3)$, integramos respecto a x agrupando por partes:

$$F = \int e^{3x}\left[(3x^2 + 2x)y + y^3\right] dx$$

Para $\int (3x^2+2x)e^{3x}\,dx$, observamos que $(x^2e^{3x})' = 2xe^{3x}+3x^2e^{3x} = (3x^2+2x)e^{3x}$, por lo que:

$$\int (3x^2 + 2x)e^{3x}\,dx = x^2e^{3x}$$

Para $\int y^3e^{3x}\,dx = \dfrac{y^3e^{3x}}{3}$.
Entonces:

$$F = e^{3x}\left(x^2y + \frac{y^3}{3}\right) + h(y)$$

Paso 5: Determinar $h(y)$ usando $F_y = N^*$.

$$\frac{\partial F}{\partial y} = e^{3x}\left(x^2 + y^2\right) + h'(y) = N^* = e^{3x}(x^2 + y^2)$$

$$h'(y) = 0 \implies h(y) = 0$$

Paso 6: Solución general.

$$\boxed{e^{3x}\left(x^2y + \frac{y^3}{3}\right) = C}$$

Verificación. Derivando implícitamente respecto a x:

$$3e^{3x}\left(x^2y + \frac{y^3}{3}\right) + e^{3x}\left(2xy + (x^2 + y^2)y'\right) = 0$$

Dividiendo entre e^{3x}:

$$3x^2y + y^3 + 2xy + (x^2 + y^2)y' = 0$$

$$(3x^2y + 2xy + y^3) + (x^2 + y^2)y' = 0\checkmark$$

Ejercicio 22 (pág. 77, [2])

Enunciado

Resolver la siguiente ecuación diferencial usando el factor integrante dado:

$$(3xy + y^2) + (x^2 + xy)\,y' = 0, \qquad \mu(x,y) = \frac{1}{xy(2x+y)}$$

Desarrollo y Respuesta

Paso 1: Verificar que la ecuación original NO es exacta.
Identificamos $M = 3xy + y^2$ y $N = x^2 + xy$:

$$\frac{\partial M}{\partial y} = 3x + 2y \qquad \frac{\partial N}{\partial x} = 2x + y$$

Como $3x + 2y \neq 2x + y$, la ecuación **no es exacta**.

Paso 2: Multiplicar por el factor integrante $\mu = \dfrac{1}{xy(2x+y)}$.

$$\frac{3xy + y^2}{xy(2x+y)} + \frac{x^2 + xy}{xy(2x+y)}\, y' = 0$$

Simplificando cada término:

$$M^* = \frac{y(3x+y)}{xy(2x+y)} = \frac{3x+y}{x(2x+y)}$$

$$N^* = \frac{x(x+y)}{xy(2x+y)} = \frac{x+y}{y(2x+y)}$$

Paso 3: Verificar exactitud de la ecuación modificada.

$$\frac{\partial M^*}{\partial y} = \frac{-1}{(2x+y)^2} \qquad \frac{\partial N^*}{\partial x} = \frac{-1}{(2x+y)^2}$$

Como son iguales, la ecuación modificada **es exacta**. ✓

Paso 4: Encontrar $F(x, y)$ integrando N^* respecto a y.

$$F = \int \frac{x+y}{y(2x+y)}\, dy$$

Descomponemos en fracciones parciales respecto a y:

$$\frac{x+y}{y(2x+y)} = \frac{1}{2} \cdot \frac{1}{y} + \frac{1}{2} \cdot \frac{1}{2x+y}$$

Integrando:

$$F = \frac{1}{2} \ln|y| + \frac{1}{2} \ln|2x+y| + g(x) = \frac{1}{2} \ln|y(2x+y)| + g(x)$$

Paso 5: Determinar $g(x)$ usando $F_x = M^*$.

$$\frac{\partial F}{\partial x} = \frac{1}{2} \cdot \frac{2}{2x+y} + g'(x) = \frac{1}{2x+y} + g'(x)$$

Igualando a $M^* = \dfrac{3x+y}{x(2x+y)}$:

$$\frac{1}{2x+y} + g'(x) = \frac{3x+y}{x(2x+y)}$$

$$g'(x) = \frac{3x+y}{x(2x+y)} - \frac{1}{2x+y} = \frac{3x+y-x}{x(2x+y)} = \frac{2x}{x(2x+y)} \cdot \frac{1}{1}$$

Simplificando con fracciones parciales:

$$g'(x) = \frac{1}{x} \implies g(x) = \ln|x|$$

Paso 6: Solución general.

$$F = \ln|x| + \frac{1}{2}\ln|y(2x+y)| = C$$

Multiplicando por 2 y aplicando propiedades del logaritmo:

$$\ln\left|x^2y(2x+y)\right| = C$$

$$\boxed{x^2y(2x+y) = C}$$

Verificación (computacional). Se confirmó mediante cálculo simbólico que $\frac{\partial M^*}{\partial y} = \frac{\partial N^*}{\partial x} = \frac{-1}{(2x+y)^2}$, garantizando la exactitud de la ecuación modificada. ✓

Lectura, Escritura y Exposición:

Actividad 4. : *¿Juega Dios a Los Dados?*: La Nueva Matemática Del Caos. Realizar una lectura, un resumen y prepare una crítica escrita de los capítulos 4 y 5 de [4].

Resumen — Capítulo 4

Resumen Capítulo 4

El capítulo 4 gira en torno a la probabilidad y cómo esta se comporta en experimentos repetidos. El **Teorema de los Grandes Números** de Bernoulli explica que a medida que más veces se realiza una acción, el resultado tiende a acercarse a su probabilidad teórica. Por ejemplo, si lanzamos una moneda pocas veces puede salir cualquier cosa, pero si la lanzamos miles de veces, la proporción de caras y sellos tenderá a ser **50/50**.

Algo importante que el capítulo toca, y que ya se había mencionado antes, es que las monedas y los dados **no son verdaderamente aleatorios**. Lo que sí es aleatorio son los factores externos que la persona no controla: la fuerza con la que se lanza, el material donde cae, el ángulo, entre otros. Si esos factores se controlaran perfectamente, el resultado siempre sería el mismo.

En cuanto a la **distribución binomial**, el capítulo muestra que aunque es posible sacar 10 caras seguidas, esto es extremadamente improbable. **Posible no es lo mismo que probable**, y a medida que aumentan los lanzamientos, los resultados siempre tenderán al centro.

Finalmente, sobre la "justicia" de una moneda, la aleatoriedad humana no viene solo de qué lado se coloca la moneda antes de lanzar, sino también de la fuerza, el ángulo y otros factores difíciles de controlar. **Todo eso en conjunto es lo que hace que el resultado sea impredecible**, no la moneda en sí misma.

Resumen — Capítulo 5

Resumen Capítulo 5

El capítulo 5 trata sobre cómo nació la **estadística** a partir de un problema real: los astrónomos tenían demasiadas observaciones y no sabían cómo usarlas todas. Tener más ecuaciones que incógnitas complicaba todo, porque cada observación tiene pequeños errores y al unir ecuaciones que dependen de otras, los errores se acumulan y las ecuaciones terminan contradiciéndose entre sí. **Euler**, siendo uno de los matemáticos más grandes de la historia, no pudo resolverlo porque intentó hacerlo de manera exacta, lo cual era imposible con datos imperfectos.

La solución la encontró **Mayer**, un astrónomo, no un matemático puro. Su idea fue simplemente agrupar las observaciones y promediarlas. Esto tiene sentido porque los errores tienden a ser igual de grandes para ambos lados, entonces al promediar **se cancelan entre sí** y el resultado se acerca bastante a la verdad. Esto muestra que a veces la intuición práctica ayuda más que el conocimiento técnico, especialmente cuando algo es demasiado complejo para resolverlo de

manera rigurosa.

Sobre la **distribución normal**, los errores no tienden a ser ni muy pequeños ni muy grandes — están en el medio. Las dos colas de la campana representan errores grandes, una hacia arriba (mediste de más) y otra hacia abajo (mediste de menos), y ambas son poco probables. Esto es útil porque **te dice qué tan confiable es tu resultado** sin necesidad de eliminar el error completamente.

Finalmente, **Kolmogorov** resolvió el debate filosófico sobre qué es la probabilidad de forma muy práctica: en lugar de definirla, escribió las reglas que debe cumplir. Sabiendo cómo se comporta se pueden hacer muchas cosas con ella, e incluso definirla a partir de eso. Con esas reglas fue suficiente para construir toda la teoría matemática moderna.

Crítica escrita

Crítica — Fancho

De los dos capítulos, el que me pareció **más importante fue el 5**, específicamente la idea de promediar observaciones imperfectas para obtener resultados confiables. Esto me parece más aplicable que el caso de las monedas y los dados, y además lo del determinismo de las monedas ya había sido mencionado en capítulos anteriores, por lo que no fue algo sorprendente.

En cuanto a **conexiones con el curso**, el capítulo 5 me recordó algo que el profesor había mencionado sobre el error en las mediciones — que siempre tiene cierta magnitud y que al promediar varias observaciones el resultado se acerca al valor real, precisamente porque los errores de ambos lados se cancelan. Por otro lado, lo del capítulo 4 sobre monedas y dados se conecta con los **sistemas dinámicos** vistos en clase: algo que parece aleatorio pero que en realidad es una función que depende de muchas condiciones iniciales que no controlamos.

Como **reflexión personal**, estos capítulos no cambiaron drásticamente mi forma de ver el azar, ya que los capítulos anteriores del libro habían ido construyendo esas mismas ideas poco a poco. Para cuando llegué aquí, las conclusiones eran casi esperables. Eso no es necesariamente malo — muestra que el libro tiene una **estructura acumulativa bien pensada**.

En cuanto al **estilo del libro**, se nota claramente que está escrito para personas que no son expertas en matemáticas. Cada capítulo plantea una idea central, la explica con ejemplos cotidianos y la refuerza con anécdotas históricas. Sin embargo, el capítulo 5 sí se pone más técnico que el 4, especialmente cuando aparecen conceptos como mínimos cuadrados o la raíz cuadrada de n, y ahí es donde se pierde un poco el ritmo, más aún si uno lo está siguiendo en audio donde las fórmulas no se pueden ver.

Problemas No Rutinarios:

Se espera que las siguientes actividades le permita examinar, manipular y aplicar apropiadamente los conceptos que se han aprendiendo en el curso, como también, identificar razones y causas matemáticas para resolverlas.

Actividad 5. Realizar los ejercicios del 17 al 30 de la página 34 de [2], entregar 18, 20, 21, 22, 26 y 28.

Ejercicio 18 (pág. 34, [2])

Enunciado

[Transcribir el enunciado del ejercicio 18 aquí.]

Desarrollo y Respuesta

Ejercicio 20 (pág. 34, [2])

Enunciado

[Transcribir el enunciado del ejercicio 20 aquí.]

Desarrollo y Respuesta

Ejercicio 21 (pág. 34, [2])

Enunciado

[Transcribir el enunciado del ejercicio 21 aquí.]

Desarrollo y Respuesta

Ejercicio 22 (pág. 34, [2])

Enunciado

[Transcribir el enunciado del ejercicio 22 aquí.]

Desarrollo y Respuesta

Ejercicio 26 (pág. 34, [2])

Enunciado

[Transcribir el enunciado del ejercicio 26 aquí.]

Desarrollo y Respuesta

Ejercicio 28 (pág. 34, [2])

Enunciado

[Transcribir el enunciado del ejercicio 28 aquí.]

Desarrollo y Respuesta

Actividad 6. Realizar los ejercicios del 17 al 31 de la página 40 de [2], entregar 20, 25, 28 y 30.

Ejercicio 20 (pág. 40, [2])

Enunciado

Resolver el siguiente problema con valor inicial:

$$y' = \frac{2 - e^x}{3 + 2y}, \qquad y(0) = 0$$

y determinar dónde la solución alcanza su valor máximo.

Desarrollo y Respuesta

Paso 1: Identificar el tipo de ecuación.
La ecuación se puede escribir como:

$$(3 + 2y)\, dy = (2 - e^x)\, dx$$

Como el lado izquierdo depende **solo de** y y el derecho **solo de** x, la ecuación es **separable**.

Paso 2: Separar e integrar ambos lados.

$$\int (3 + 2y)\, dy = \int (2 - e^x)\, dx$$

$$3y + y^2 = 2x - e^x + C$$

Paso 3: Aplicar la condición inicial $y(0) = 0$**.**
Sustituimos $x = 0$, $y = 0$:

$$3(0) + (0)^2 = 2(0) - e^0 + C \implies 0 = -1 + C \implies C = 1$$

La solución implícita del PVI es:

$$\boxed{y^2 + 3y = 2x - e^x + 1}$$

Paso 4: Encontrar el máximo.
La solución alcanza su máximo cuando $y' = 0$, es decir:

$$\frac{2 - e^x}{3 + 2y} = 0 \implies 2 - e^x = 0 \implies x = \ln 2$$

Sustituimos $x = \ln 2$ en la solución para hallar y:

$$y^2 + 3y = 2\ln 2 - e^{\ln 2} + 1 = 2\ln 2 - 2 + 1 = 2\ln 2 - 1$$

$$y^2 + 3y - (2\ln 2 - 1) = 0$$

Usando la fórmula cuadrática:

$$y = \frac{-3 \pm \sqrt{9 + 4(2\ln 2 - 1)}}{2} = \frac{-3 \pm \sqrt{5 + 8\ln 2}}{2}$$

Tomando la raíz positiva (consistente con $y(0) = 0$):

$$y_{\text{máx}} = \frac{-3 + \sqrt{5 + 8\ln 2}}{2} \approx 0{,}1237$$

El máximo ocurre en $x = \ln 2 \approx 0{,}693$.

Verificación. Diferenciando implícitamente $y^2 + 3y = 2x - e^x + 1$ respecto a x:

$$(2y + 3)\, y' = 2 - e^x \implies y' = \frac{2 - e^x}{3 + 2y} \checkmark$$

Esto coincide exactamente con la ecuación original.

Ejercicio 25 (pág. 40, [2])

Enunciado

Considerar la ecuación:

$$\frac{dy}{dx} = \frac{y - 4x}{x - y} \tag{29}$$

a. Mostrar que la ecuación es homogénea reescribiéndola como función de y/x.

b. Introducir $v = y/x$, expresar dy/dx en términos de x, v y dv/dx.

c. Mostrar que la ecuación resultante es separable.

d. Resolver la ecuación separable obteniendo v implícitamente en términos de x.

e. Reemplazar v por y/x para obtener la solución.

f. Graficar el campo de direcciones y algunas curvas integrales.

Desarrollo y Respuesta

a. Verificar que la ecuación es homogénea.
Reescribimos dividiendo numerador y denominador entre x:

$$\frac{dy}{dx} = \frac{y - 4x}{x - y} = \frac{(y/x) - 4}{1 - (y/x)} = F\left(\frac{y}{x}\right)$$

Como el lado derecho depende **únicamente** del cociente y/x, la ecuación es **homogénea**. ✓

b. Sustitución $v = y/x$.

Sea $v = y/x$, es decir, $y = vx$. Derivando respecto a x:

$$\frac{dy}{dx} = v + x\frac{dv}{dx}$$

c. Ecuación separable resultante.
Sustituimos en la ecuación original:

$$v + x\frac{dv}{dx} = \frac{v-4}{1-v}$$

$$x\frac{dv}{dx} = \frac{v-4}{1-v} - v = \frac{v-4-v(1-v)}{1-v} = \frac{v-4-v+v^2}{1-v} = \frac{v^2-4}{1-v}$$

La ecuación resultante es:

$$x\frac{dv}{dx} = \frac{v^2-4}{1-v}$$

que es **separable** porque las variables v y x se pueden separar.

d. Resolver la ecuación separable.
Separando variables:

$$\frac{1-v}{v^2-4}\,dv = \frac{dx}{x}$$

Descomponemos en fracciones parciales con $v^2 - 4 = (v-2)(v+2)$:

$$\frac{1-v}{(v-2)(v+2)} = \frac{A}{v-2} + \frac{B}{v+2}$$

$$1 - v = A(v+2) + B(v-2)$$

$$v = 2: \quad -1 = 4A \implies A = -\tfrac{1}{4} \qquad v = -2: \quad 3 = -4B \implies B = -\tfrac{3}{4}$$

Integrando:

$$\int\left(\frac{-1/4}{v-2} + \frac{-3/4}{v+2}\right)dv = \int\frac{dx}{x}$$

$$-\tfrac{1}{4}\ln|v-2| - \tfrac{3}{4}\ln|v+2| = \ln|x| + C_1$$

Multiplicando por -4:

$$\ln|v-2| + 3\ln|v+2| = -4\ln|x| + C$$

$$\ln\left|(v-2)(v+2)^3\right| = \ln\left|x^{-4}\right| + C$$

$$(v-2)(v+2)^3 = \frac{C}{x^4}$$

e. Solución en términos de x e y.
Reemplazando $v = y/x$:

$$\left(\frac{y}{x} - 2\right)\left(\frac{y}{x} + 2\right)^3 = \frac{C}{x^4}$$

$$\frac{y-2x}{x}\cdot\frac{(y+2x)^3}{x^3} = \frac{C}{x^4}$$

$$\boxed{(y-2x)(y+2x)^3 = C}$$

e. Solución en términos de x e y.
Reemplazando $v = y/x$:

$$\left(\frac{y}{x}-2\right)\left(\frac{y}{x}+2\right)^3 = \frac{C}{x^4}$$

$$\frac{y-2x}{x}\cdot\frac{(y+2x)^3}{x^3} = \frac{C}{x^4}$$

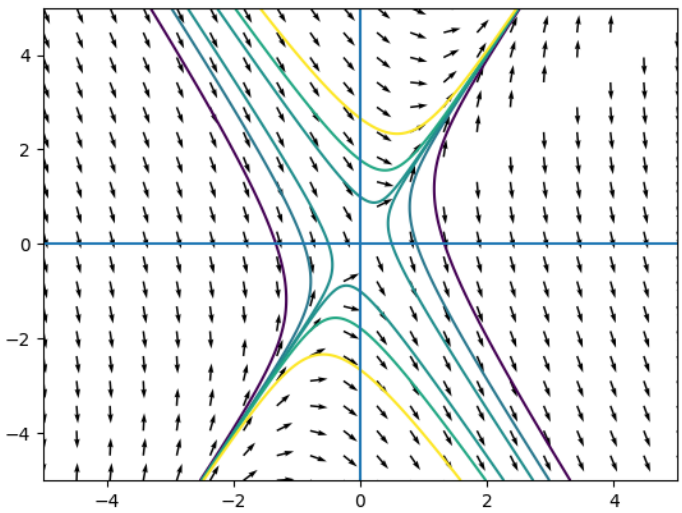
$$\boxed{(y-2x)(y+2x)^3 = C}$$

f. Campo de direcciones y curvas integrales.
La gráfica fue generada externamente (Python) evaluando

$$\frac{dy}{dx} = \frac{y-4x}{x-y}$$

en una malla de puntos y superponiendo curvas de nivel de la solución implícita

$$(y-2x)(y+2x)^3 = C.$$



La simetría respecto al origen se verifica ya que

$$(y-2x)(y+2x)^3 \xrightarrow{(x,y)\to(-x,-y)} (-y+2x)(-y-2x)^3 = (y-2x)(y+2x)^3$$

Ejercicio 28 (pág. 40, [2])

Enunciado

Para la siguiente ecuación:

a. Mostrar que es homogénea.

b. Resolver la ecuación.

c. Graficar el campo de direcciones y algunas curvas integrales. ¿Son simétricas respecto al origen?

$$\frac{dy}{dx} = \frac{4y - 3x}{2x - y}$$

Desarrollo y Respuesta

a. Verificar homogeneidad.
Dividiendo numerador y denominador entre x:

$$\frac{dy}{dx} = \frac{4y - 3x}{2x - y} = \frac{4(y/x) - 3}{2 - (y/x)} = F\Big(\frac{y}{x}\Big)$$

La expresión depende únicamente del cociente y/x, por lo tanto la ecuación es **homogénea**. ✓

b. Resolver la ecuación.
Sustitución: Sea $v = y/x$, es decir, $y = vx$, de donde $\dfrac{dy}{dx} = v + x\dfrac{dv}{dx}$.
Sustituyendo:

$$v + x\frac{dv}{dx} = \frac{4v - 3}{2 - v}$$

$$x\frac{dv}{dx} = \frac{4v - 3}{2 - v} - v = \frac{4v - 3 - v(2 - v)}{2 - v} = \frac{v^2 + 2v - 3}{2 - v} = \frac{(v + 3)(v - 1)}{2 - v}$$

Separando variables:

$$\frac{2 - v}{(v + 3)(v - 1)}\, dv = \frac{dx}{x}$$

Fracciones parciales:

$$\frac{2 - v}{(v + 3)(v - 1)} = \frac{A}{v + 3} + \frac{B}{v - 1}$$

$$2 - v = A(v - 1) + B(v + 3)$$

$$v = 1: \quad 1 = 4B \implies B = \tfrac{1}{4} \qquad v = -3: \quad 5 = -4A \implies A = -\tfrac{5}{4}$$

Integrando:

$$\int \left(\frac{-5/4}{v + 3} + \frac{1/4}{v - 1}\right) dv = \int \frac{dx}{x}$$

$$-\tfrac{5}{4}\ln|v + 3| + \tfrac{1}{4}\ln|v - 1| = \ln|x| + C_1$$

Multiplicando por 4:

$$\ln|v-1| - 5\ln|v+3| = 4\ln|x| + C$$

$$\ln\left|\frac{v-1}{(v+3)^5}\right| = \ln\left|x^4\right| + C$$

$$\frac{v-1}{(v+3)^5} = Cx^4$$

Reemplazando $v = y/x$:

$$\frac{y/x-1}{(y/x+3)^5} = Cx^4 \qquad \Longrightarrow \qquad \frac{(y-x)/x}{(y+3x)^5/x^5} = Cx^4$$

$$\frac{(y-x)\,x^4}{(y+3x)^5} = Cx^4$$

$$\boxed{\frac{y-x}{(y+3x)^5} = C}$$

c. Campo de direcciones y curvas integrales.
Evaluando $dy/dx = (4y-3x)/(2x-y)$ en una malla se obtiene el campo de direcciones. Las curvas integrales corresponden a diferentes valores de C. Por ser la ecuación homogénea, las curvas son **simétricas respecto al origen**.

Ejercicio 30 (pág. 40, [2])

Enunciado

Para la siguiente ecuación:

a. Mostrar que es homogénea.

b. Resolver la ecuación.

c. Graficar el campo de direcciones y algunas curvas integrales. ¿Son simétricas respecto al origen?

$$\frac{dy}{dx} = \frac{x^2-3y^2}{2xy}$$

Desarrollo y Respuesta

a. Verificar homogeneidad.
Dividiendo numerador y denominador entre x^2:

$$\frac{dy}{dx} = \frac{x^2-3y^2}{2xy} = \frac{1-3(y/x)^2}{2(y/x)} = F\left(\frac{y}{x}\right)$$

La expresión depende únicamente del cociente y/x, por lo tanto la ecuación es **homogénea**. ✓

b. Resolver la ecuación.
Sustitución: Sea $v = y/x$, es decir, $y = vx$, de donde $\dfrac{dy}{dx} = v + x\dfrac{dv}{dx}$.
Sustituyendo:

$$v + x\frac{dv}{dx} = \frac{1-3v^2}{2v}$$

$$x\frac{dv}{dx} = \frac{1-3v^2}{2v} - v = \frac{1-3v^2-2v^2}{2v} = \frac{1-5v^2}{2v}$$

Separando variables:

$$\frac{2v}{1-5v^2}\,dv = \frac{dx}{x}$$

Integrando el lado izquierdo con $u = 1-5v^2$, $du = -10v\,dv$:

$$\int \frac{2v}{1-5v^2}\,dv = -\frac{1}{5}\ln\left|1-5v^2\right|$$

Entonces:

$$-\frac{1}{5}\ln|1-5v^2| = \ln|x| + C_1$$

$$\ln|1-5v^2| = -5\ln|x| + C$$

$$|1-5v^2| = \frac{C}{x^5}$$

Reemplazando $v = y/x$:

$$1 - 5\frac{y^2}{x^2} = \frac{C}{x^5}$$

$$\frac{x^2-5y^2}{x^2} = \frac{C}{x^5}$$

$$x^5(x^2-5y^2) = C\,x^2$$

$$\boxed{x^3(x^2-5y^2) = C}$$

c. Campo de direcciones y curvas integrales.
Evaluando $dy/dx = (x^2-3y^2)/(2xy)$ en una malla se obtiene el campo de direcciones. Las curvas $x^3(x^2-5y^2) = C$ para distintos valores de C forman las curvas integrales. Por ser la ecuación homogénea, las curvas son **simétricas respecto al origen**.

Activity 7. **Computer y submit the exercises from 24 to 27 of the page 60 of [2]**

Bernoulli Equations. Sometimes it is possible to solve a nonlinear equation by making a change of the dependent variable that converts it into a linear equation. The most important such equation has the form

$$y' + p(t)y = q(t)y^n,$$

and is called a Bernoulli equation after Jakob Bernoulli. Problems 23 and 25 deal with equations of this type.

24. $y' = ry - ky^2$**,** $r > 0$ **and** $k > 0$**. This equation is important in population dynamics and is discussed in detail in Section 2.5.**

Desarrollo y Respuesta

Theorem 1. *This part was so fucking hard, so I used heavy IA to solve it, as well the time for this assignment wasn't enough, I didn't put enough time for learning the real purpose of the "Bernoulli"stuff, good reading anyway.*

This is a Bernoulli equation with $p(t) = -r$, $q(t) = -k$, and $n = 2$.
Rewrite in standard Bernoulli form.

$$y' - ry = -ky^2$$

Apply the Bernoulli substitution.
Divide both sides by $y^n = y^2$:

$$y^{-2}y' - ry^{-1} = -k$$

Let $v = y^{1-n} = y^{-1}$, so that $v' = -y^{-2}y'$, which gives $y^{-2}y' = -v'$.
Substituting:

$$-v' - rv = -k$$

$$v' + rv = k$$

Solve the linear equation.
This is a first-order linear ODE in v. The integrating factor is:

$$\mu(t) = e^{\int r\,dt} = e^{rt}$$

Multiplying both sides by $\mu(t)$:

$$\frac{d}{dt}\left(e^{rt}v\right) = ke^{rt}$$

Integrating both sides:

$$e^{rt}v = \frac{k}{r}e^{rt} + C$$

$$v = \frac{k}{r} + Ce^{-rt}$$

Back-substitute $v = y^{-1}$.

$$\frac{1}{y} = \frac{k}{r} + Ce^{-rt}$$

$$\boxed{y = \frac{1}{\frac{k}{r} + Ce^{-rt}}}$$

Analyze the behavior.
As $t \to \infty$, since $r > 0$ we have $e^{-rt} \to 0$, so:

$$y \to \frac{1}{k/r} = \frac{r}{k}$$

This confirms that the population approaches the carrying capacity $\frac{r}{k}$ regardless of the initial condition (provided $y(0) > 0$), which is the hallmark of logistic growth.

25. $y' = \epsilon y - \sigma y^3$**,** $\epsilon > 0$ **and** $\sigma > 0$**.**

Desarrollo y Respuesta

This is a Bernoulli equation with $p(t) = -\epsilon$, $q(t) = -\sigma$, and $n = 3$.
Standard Bernoulli form.

$$y' - \epsilon y = -\sigma y^3$$

Bernoulli substitution.
Divide both sides by $y^n = y^3$:

$$y^{-3}y' - \epsilon y^{-2} = -\sigma$$

Let $v = y^{1-n} = y^{-2}$, so that $v' = -2y^{-3}y'$, which gives $y^{-3}y' = -\frac{v'}{2}$.
Substituting:

$$-\frac{v'}{2} - \epsilon v = -\sigma$$

Multiplying through by -2:

$$v' + 2\epsilon v = 2\sigma$$

Solving the linear equation.
This is a first-order linear ODE in v. The integrating factor is:

$$\mu(t) = e^{\int 2\epsilon\, dt} = e^{2\epsilon t}$$

Multiplying both sides by $\mu(t)$:

$$\frac{d}{dt}\left(e^{2\epsilon t}v\right) = 2\sigma e^{2\epsilon t}$$

Integrating both sides:

$$e^{2\epsilon t}v = \frac{\sigma}{\epsilon}e^{2\epsilon t} + C$$

$$v = \frac{\sigma}{\epsilon} + Ce^{-2\epsilon t}$$

Back-substitution $v = y^{-2}$.

$$\frac{1}{y^2} = \frac{\sigma}{\epsilon} + Ce^{-2\epsilon t}$$

$$\boxed{y = \frac{1}{\sqrt{\frac{\sigma}{\epsilon} + Ce^{-2\epsilon t}}}}$$

Long-term behavior.
As $t \to \infty$, since $\epsilon > 0$ we have $e^{-2\epsilon t} \to 0$, so:

$$y \to \frac{1}{\sqrt{\sigma/\epsilon}} = \sqrt{\frac{\epsilon}{\sigma}}$$

Thus the solution approaches the equilibrium value $\sqrt{\epsilon/\sigma}$ as $t \to \infty$, indicating that small perturbations in the fluid flow decay toward this stable state.

Discontinuous Coefficients. Linear differential equations sometimes occur in which one or both of the functions p and g have jump discontinuities. If t_0 is such a point of discontinuity, then it is necessary to solve the equation separately for $t < t_0$ and $t > t_0$. Afterward, the two solutions are matched so that y is continuous at t_0; this is accomplished by a proper choice of the arbitrary constants. The following two problems illustrate this situation. Note in each case that it is impossible also to make y' continuous at t_0.

26. Solve the initial value problem

$$y' + 2y = g(t), \qquad y(0) = 0,$$

where

$$g(t) = \begin{cases} 1, & 0 \le t \le 1, \\ 0, & t > 1. \end{cases}$$

Desarrollo y Respuesta

The function $g(t)$ has a jump discontinuity at $t_0 = 1$, so we solve the equation separately on $[0, 1]$ and $(1, \infty)$, then match the solutions by requiring y to be continuous at $t = 1$.
Integrating factor.
For both intervals the left-hand side has the same form. The integra-

ting factor is:

$$\mu(t) = e^{\int 2\,dt} = e^{2t}$$

so the equation becomes in each interval:

$$\frac{d}{dt}\left(e^{2t}y\right) = g(t)e^{2t}$$

Solution on $0 \le t \le 1$.
Here $g(t) = 1$, so:

$$\frac{d}{dt}\left(e^{2t}y\right) = e^{2t}$$

$$e^{2t}y = \frac{1}{2}e^{2t} + C_1$$

$$y_1 = \frac{1}{2} + C_1 e^{-2t}$$

Applying $y(0) = 0$:

$$0 = \frac{1}{2} + C_1 \implies C_1 = -\frac{1}{2}$$

$$y_1 = \frac{1}{2}\left(1 - e^{-2t}\right), \qquad 0 \le t \le 1$$

Solution on $t > 1$.
Here $g(t) = 0$, so:

$$\frac{d}{dt}\left(e^{2t}y\right) = 0$$

$$e^{2t}y = C_2$$

$$y_2 = C_2 e^{-2t}, \qquad t > 1$$

Matching at $t = 1$.
We require y to be continuous at $t = 1$, i.e. $y_1(1) = y_2(1)$:

$$\frac{1}{2}\left(1 - e^{-2}\right) = C_2 e^{-2}$$

$$C_2 = \frac{e^2 - 1}{2}$$

Complete solution.

$$\boxed{y = \begin{cases} \frac{1}{2}(1 - e^{-2t}), & 0 \le t \le 1, \\ \frac{e^2 - 1}{2}\, e^{-2t}, & t > 1. \end{cases}}$$

Remark. Although y is continuous at $t = 1$, the derivative y' is not: on the left $y'(1^-) = e^{-2}$, while on the right $y'(1^+) = -(e^2 - 1)e^{-2}$, confirming the observation made in the problem statement.

27. Solve the initial value problem

$$y' + p(t)y = 0, \qquad y(0) = 1,$$

where

$$p(t) = \begin{cases} 2, & 0 \le t \le 1, \\ 1, & t > 1. \end{cases}$$

Desarrollo y Respuesta

The function $p(t)$ has a jump discontinuity at $t_0 = 1$, so we solve separately on $[0, 1]$ and $(1, \infty)$, then match at $t = 1$ by requiring continuity of y.

Solution on $0 \le t \le 1$.

Here $p(t) = 2$, so the equation is $y' + 2y = 0$. The integrating factor is $\mu(t) = e^{2t}$, giving:

$$\frac{d}{dt}\left(e^{2t}y\right) = 0 \implies e^{2t}y = C_1 \implies y_1 = C_1 e^{-2t}$$

Applying $y(0) = 1$:

$$1 = C_1 \implies C_1 = 1$$

$$y_1 = e^{-2t}, \qquad 0 \le t \le 1$$

Solution on $t > 1$.

Here $p(t) = 1$, so the equation is $y' + y = 0$. The integrating factor is $\mu(t) = e^t$, giving:

$$\frac{d}{dt}\left(e^{t}y\right) = 0 \implies e^{t}y = C_2 \implies y_2 = C_2 e^{-t}$$

Matching at $t = 1$.

We require $y_1(1) = y_2(1)$:

$$e^{-2} = C_2 e^{-1} \implies C_2 = e^{-1}$$

Complete solution.

$$\boxed{y = \begin{cases} e^{-2t}, & 0 \le t \le 1, \\ e^{-1}\, e^{-t}, & t > 1. \end{cases}}$$

which can be written more compactly for $t > 1$ as $y = e^{-(t+1)}$.

Remark. As expected, y is continuous at $t = 1$ since both pieces give $y(1) = e^{-2}$, but y' is not: on the left $y'(1^-) = -2e^{-2}$, while on the right $y'(1^+) = -e^{-2}$, reflecting the jump in $p(t)$ at that point.

Análisis Numérico o Computacional:

Actividad 8. **(Entregar).** [Mix Numérico – No Rutinario] Considere el problema con valor inicial:

$$\frac{dy}{dt} + \left(\sqrt{1+\sin^2(t)}\right) y = t, \qquad y(0) = 2$$

1. Utilice la definición de integral o el Teorema Fundamental del Cálculo para deducir que el factor integrante para esta ecuación diferencial se puede escribir como:
$$\mu(t) = e^{\int_0^t \sqrt{1+\sin^2(s)}\, ds};$$
y que la solución del problema con valor inicial es:
$$y(t) = \frac{1}{\mu(t)} \int_0^t s\, \mu(s)\, ds + \frac{2}{\mu(t)}.$$

Punto 1 — Factor Integrante

Dada la ecuación:
$$\frac{dy}{dt} + \sqrt{1+\sin^2(t)}\, y = t,$$
se identifica como una ecuación lineal de la forma:
$$y' + a(t)y = b(t),$$
donde $a(t) = \sqrt{1+\sin^2(t)}$.
El factor integrante está dado por:
$$\mu(t) = e^{\int_0^t \sqrt{1+\sin^2(s)}\, ds}.$$
Multiplicando toda la ecuación por $\mu(t)$, se obtiene:
$$\mu(t)\frac{dy}{dt} + \mu(t)\sqrt{1+\sin^2(t)}\, y = t\mu(t).$$
El lado izquierdo corresponde a la derivada del producto:
$$\frac{d}{dt}\big(\mu(t)y(t)\big) = t\mu(t).$$
Integrando en el intervalo $[0, t]$:
$$\mu(t)y(t) - \mu(0)y(0) = \int_0^t s\, \mu(s)\, ds.$$
Como $\mu(0) = 1$ y $y(0) = 2$, se tiene:
$$\mu(t)y(t) = \int_0^t s\, \mu(s)\, ds + 2.$$
Finalmente, despejando $y(t)$:
$$y(t) = \frac{1}{\mu(t)} \int_0^t s\, \mu(s)\, ds + \frac{2}{\mu(t)}.$$

2. Obtenga una aproximación de la solución en $t = 1$. Utilice un método de integración numérica (como la regla de Simpson) en un ciclo anidado para estimar valores de $\mu(t)$, y con ello, estimar el valor de $\displaystyle\int_0^1 s\,\mu(s)\,ds$.

   [Sugerencia]: primero utilice el método de integración numérica para estimar $\mu(t)$ en $t = 0{,}1, 0{,}2, 0{,}3, \ldots, 1$. Luego, utilice estos valores y aplique de nuevo el método de integración numérica para aproximar $\displaystyle\int_0^1 s\,\mu(s)\,ds$.

Punto 2 — Aproximación Numérica

Se utilizó la regla de Simpson compuesta con paso $h = 0{,}1$.

1. Tabla de valores de $\mu(t)$

t	$\mu(t)$
0.1	...
0.2	...
0.3	...
0.4	...
0.5	...
0.6	...
0.7	...
0.8	...
0.9	...
1.0	...

2. Aproximación de la integral

$$\int_0^1 s\mu(s)\,ds \approx \text{(valor numérico)}$$

3. Valor aproximado de $y(1)$

$$y(1) \approx \text{(valor final)}$$

Implementación en Python

```
import math

def simpson(f, a, b, n):
    h = (b - a) / n
    s = f(a) + f(b)
    for i in range(1, n):
        s += 4*f(a + i*h) if i % 2 else 2*f(a + i*h)
    return s * h / 3

def f(s):
    return math.sqrt(1 + math.sin(s)**2)
```

```
def mu(t):
    return math.exp(simpson(f, 0, t, 100))

ts = [i/10 for i in range(1, 11)]
mu_vals = [mu(t) for t in ts]

print("t | mu(t)")
for t, m in zip(ts, mu_vals):
    print(f"{t:.1f} | {m:.6f}")

def g_index(i):
    return ts[i] * mu_vals[i]

h = 0.1
S = g_index(0) + g_index(9)
for i in range(1, 9):
    S += 4*g_index(i) if i % 2 else 2*g_index(i)

I = S * h / 3
y1 = I / mu_vals[-1] + 2 / mu_vals[-1]

print("\nIntegral ~", I)
print("y(1) ~", y1)
```

3. Utilice el método de Euler para aproximar la solución en $t = 1$, con tamaños de paso $h = 0{,}1$ y $h = 0{,}01$.

Punto 3 — Método de Euler

Se utiliza el método de Euler:

$$y_{n+1} = y_n + h\, f(t_n, y_n), \quad f(t, y) = t - \sqrt{1 + \sin^2(t)}\, y.$$

Euler con $h = 0{,}1$:

n	t_n	y_n
0	0.0	2.0
1	0.1	$y_1 = 2 + 0{,}1\, f(0, 2)$
2	0.2	$y_2 = y_1 + 0{,}1\, f(0{,}1, y_1)$
$\vdots$	$\vdots$	$\vdots$
10	1.0	...

Euler con $h = 0{,}01$:
Se repite el mismo procedimiento con $h = 0{,}01$ hasta $t = 1$.

$$y(1) \approx \text{(valor numérico)}$$

4. Como ingeniero: proponga una metodología para comparar la precisión y la eficiencia de los esquemas numéricos en 2 y 3.

Punto 4 — Comparación de Métodos

Criterios de comparación
Se consideran dos aspectos:

- **Precisión:** medida mediante el error relativo respecto a una solución de referencia (Simpson):

$$\text{Error} = \frac{|y_{\text{num}} - y_{\text{ref}}|}{|y_{\text{ref}}|}$$

- **Eficiencia:** estimada por el número de evaluaciones de la función $f(t, y)$.

Tabla comparativa en $t = 1$:

Método	$y(1)$	Error relativo	Costo (eval.)
Simpson	...	—	Alto
Euler $h = 0{,}1$	...	...	Bajo
Euler $h = 0{,}01$	...	...	Medio

Análisis
El método de Simpson presenta mayor precisión al utilizar integración de orden superior, aunque con mayor costo computacional debido al cálculo anidado.
El método de Euler con $h = 0{,}1$ es computacionalmente eficiente, pero presenta un error significativo. Al reducir el paso a $h = 0{,}01$, la precisión mejora, pero el costo aumenta proporcionalmente.
Conclusión
Existe un compromiso entre precisión y eficiencia. Simpson es preferible cuando se requiere alta precisión, mientras que Euler puede ser adecuado para estimaciones rápidas con menor costo computacional.

References

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[4] Hawking, S. *Juega Dios a los dados.* Booket Planeta (2014).